ONLINE REPRODUCIBILITY REPORT

CUTLINE: Verifying Real Product Pixels in AI-Composited Listings

Reproducibility Report

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About this report. This report accompanies the manuscript and its Supplementary Material and is published with the code at `https://github.com/yebof/cutline` (folder `reproducibility`). It holds every protocol detail, data table, figure and run note of the evaluation that the manuscript and the supplement summarise: the pre-registrations and development sweeps (§R1), one section per experiment (§R2–§R12), a worked example, the system architecture and the ledger details (§R13–§R15), the full positioning table (§R16), the previous full text of the manuscript's evaluation (§R17), the run-integrity record (§R18) and an index of the evidence files behind every table and figure (Appendix R-A). Sections, tables and figures are numbered "R*n*"; "S*n*" and the Roman-numbered tables refer to the Supplementary Material, "main" to the manuscript. Every number is taken from the experiment reports of the artifact; the claim register maps each printed number to its report. The evidence tiers `X0`, `X1`, `XP` and tier `X` of the paper are called E0, E1, EP and tier E in the code, the registrations and some figures of this report.

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R1 Pre-registration and development

R1.1 Pre-registrations

Each block was frozen in the repository before its first full run; every result line carries the hash of its pre-registration.

TABLE R1. Pre-registrations and freeze commits of the five main experiment blocks.

Block	Experiments	Freeze	Development sweeps before / after the freeze
1	E1, E2, E3, E6	commit 0387702	listed in the block's report / 0
2	E9, E10	commit 6be838f	listed / 0
3	E11	commit 039010f	DEV-1..DEV-3 / 0
4	E4	commit ec955b0 (F2)	21 / 0
5	E5, workflows, PGD, E12	commit b7bd84b (F1); prompts F0 fcc4ea7	16 / 0 (one post-F1 dev-material sweep, §R1.3)

R1.2 Registration details

Every registration is a commit of the project's private repository (timestamps UTC). The hashes are internal: no external timestamp (e.g., RFC 3161 or OpenTimestamps) was obtained.

- **Pre-registered (main blocks).** E1–E12 in the five blocks of Table R1, each frozen before its first full run.
- **Pre-registered (additional experiments).** Registered after the E1–E12 results were known, each before its own run: E13–E17 frozen together (freeze record 2026-10-07T21:41:56Z; commit 4ba86a4 at 21:42:22Z) before any run on test masters, with 0 post-registration changes (deviations: the E13 adversarial-scene arm was removed from scope during development; E15 masters are all ABO; E16 ran under load with one master per size; the E17 classifier gained the EIP-7623 floor rule before the results were final); E17b in 18afbeb (2026-10-07T23:13:27Z), re-running the rejected 65,535-MCU court proof with only Besu's transaction-pool size limit raised; E16b in 680e70e (2026-10-07T23:36:30Z), re-running E16 on a quiet host with a thread test; E18 (displayed pixels), E19 (per-pixel comparator) and E20 (cause of the G2 renditions at 500×375, from stored records) in afd30d3 (2026-10-07T23:48:06Z), each before its run or analysis; E21 (the X1-cell comparator) in 076b16d (2026-10-08T00:55:43Z), before its run. The main blocks' registrations are unchanged.
- **Post-registration analyses and fixes (disclosed).** The items of §R1.4; outcomes in Table R2.

R1.3 Registration record

Every registration is a commit of the project repository (timestamps UTC). The commits carry no external timestamp (e.g., RFC 3161 or OpenTimestamps). Table R2 lists every registration with its outcome; the post-registration changes follow. Development sweeps and amendments are in Sec. R1.6.

R1.4 Post-registration fixes, analyses and decisions

E6 fixes. Three fixes precede a single rerun with one code digest and the registered criteria and inputs (main Sec. VIII-D): (1) a chroma edge-padding model for Go's encoder (class 5), whose renditions at 1000×750 were unlinked; (2) U is computed once on the source listing's grid and inherited by renditions, instead of being re-tested on each rendition's grid; (3) the deviation metric and the hop proposition are restricted to footprint-certified pixels (Sec. S-VII). Gate E6 is reported as passing after these three fixes.

E10 fork control. The frozen script counted blocks from a poll taken before the split containers answered, so the control clause was false as written; it is recomputed from the recorded heads (side B stays at the prefix head 60), without a rerun.

E11. A finer two-scale grid of beacon delays was probed after the freeze (Fig. R11). The $(j + 2) \cdot 2$ s envelope (j censoring proposers) is an exploratory description, made after the Besu runs, of the registered inclusion bound $(j + 1) \cdot 2$ s; the registered comparison stays failed.

E4. The frozen decision code also compared 18 TrustMark rows that the scope text excluded; the verdict (0 dominations) is the same without them. Mesorch's image score is the maximum of its probability map, as frozen; its FPR-1% threshold saturates at 1.0. The registered honest tier-T coverage check fails on one thin product that has no 256-px listing, a listing-size limit; a claim that certifies nothing remains G0. The reference-holding tolerant rows were read as an equal-information comparator after the E4 results were known (no new run); the main paper uses the registered E19 instead (Sec. R4).

U out of the level criteria (after E5). Gate E5 failed, so the switch that makes U a level criterion was turned off after the run, as the pre-written consequence required. E6 and WF3, computed before the switch, were re-derived from the same

TABLE R2. Registration record. Each commit precedes the first full run (main blocks) or the experiment's own run or analysis (additional experiments); times are commit times (UTC). The registered criteria and full outcomes are in main Table II and Table R7.

Experiment	Registered / disclosed	Commit	Commit time (UTC)	Outcome
Main blocks, pre-registered before the first full run				
E1, E2, E3, E6 (block 1)	pre-registered	0387702	2026-09-30 01:22:57	E1–E3 hold (0/17,940 honest failures; 36/36 cells detected); gate E6 failed as registered and passes after three disclosed fixes (Sec. R1.4)
E9, E10 (block 2)	pre-registered	6be838f	2026-09-30 13:17:42	court equals the reference on 22,306/22,306 cases; 3/6 E10 knee predictions fail; fork control recomputed from recorded heads (disclosed)
E11 (block 3)	pre-registered	039010f	2026-09-30 08:04:03	outcome classes hold; Besu inclusion bound exceeded on 14/100 repetitions ($(k+2)$-period envelope exploratory)
E4 (block 4)	pre-registered	ec955b0	2026-10-07 00:05:33	H4 holds (0/189 dominations, 0/598 honest convictions); tier-T FRR fails on one profile (3/299); honest tier-T coverage 298/299
E5, workflows, PGD, E12 (block 5)	pre-registered (generation prompts registered earlier)	b7bd84b; prompts fcc4ea7	2026-10-07 00:09:04; prompts 2026-09-30 10:20:49	gate E5 (a1) fails, so U is printed only; PGD gate fails (surrogate white-box label); WF1, WF2 pass, WF3 one-hop only; E12 design target missed
Additional experiments, registered after the main results, each before its own run or analysis				
E13–E17	pre-registered together	4ba86a4	2026-10-07 21:42:22 (freeze record 21:41:56)	E13 holds (adversarial-scene arm not run); E14 descriptive; E15 holds; E16 partly / fails; E17 11/12 rows on Besu
E17b	registered after E17	18afbeb	2026-10-07 23:13:27	65,535-MCU proof mined with the pool limit raised
E16b	registered after E16	680e70e	2026-10-07 23:36:30	threading explanation confirmed
E18, E19, E20	registered before run / analysis	afd30d3	2026-10-07 23:48:06	E18 holds (3,600/3,600); E19 below CUTLINE on 17/18 cells; E20 met literally, wrong in substance
E21	registered before run	076b16d	2026-10-08 00:55:43	below CUTLINE on 0/18 cells (100% everywhere)
E4 reference-holding rows	post-registration analysis (disclosed)	–	–	extraction with values known; superseded by E19

records (no evaluation rerun; the stored aggregates are reproduced as a control): the post-fix E6 is unchanged; the pre-registered E6 has 58 instead of 114 honest star failures (still FAIL) and 38/38 attack cases; WF3 star ladders stay at 100%, WF3 cascades fall to about 0.5 (Tables R9, R11). The registered workflow summary computed the source levels of WF3 with U as a level criterion and those of WF1 and WF2 with U printed only. Read consistently with U printed only, under the registered rule (point share ≥ 0.5; deployability claimed only for a passing family), WF3 is deployable on one-hop (star) ladders and not on cascades; the original status is kept as the record.

PGD gate. The canonical gate failed (IoU minimum $0.976 < 0.99$), so the registered fallback labels the PGD results "surrogate white-box". The PGD preprocessing uses antialiased bilinear resizing, chosen on stand-in material before the freeze (IoU 0.9906, against 0.964 for bicubic).

E5 pilot. The 15% target, $\tau_U = 32$, the adjacency form of U (components C_{adj}), the rejection of band variants and the ground-truth validity rule were fixed on a pilot of 29 listings (design probes) before the E5 pre-registration.

E12. E12 ran twice under the same registration, the first time on a host loaded by concurrent jobs; both runs are reported (Table R19), and both miss the design target.

R1.5 Open items

Main Table II lists every registered outcome. Open items are adversarial scene content against the capture segmenter (E13, not run); Objectron and CO3D in the confirmatory E6 rerun (E15); a native verifier; an app-side coverage check; platforms that crop to a fixed aspect; and an external timestamp of the registration commits.

R1.6 Development sweeps and amendments

- **E11 (block 3).** Before the freeze: DEV-1 (8,558 items, 2,888 mismatches against the prediction, of which 141 genuine), DEV-2 and DEV-3 after harness fixes; harness fixes and two scenario changes (SC-1, SC-2) are listed in the pre-registration. After the freeze: 242 of 1,674 cells differ from the effective prediction (32 at $k \leq f$, 6 at $f+1$, 204 at $2f+1$) and 321 from the signed one; five amended predictions (AM-1..AM-5) apply only to vacuous or internally inconsistent cells, and both predictions are printed; four findings at $k \leq f$ (F1–F4) concern the comparators A–C′, not D.
- **E4 (block 4).** 21 development sweeps merged at the freeze, 0 after it; 0 amendments.
- **E5, workflows, PGD, E12 (block 5).** 16 development sweeps merged at the freeze, 0 after it; one post-freeze sweep of development material generated the product-conditioned fills of the PGD gate's development listings (development masters only); 0 amendments.

TABLE R3. Tier-X encoder classes (E1). Admitted members of the encoder zoo, family slack and held-out FRR.

Class	Name	Test	Admitted members	Need	Slack	E1 k/n	FRR_{95}	Rule 2
1	X0	EXACT: c = quant_islow(fdct_islow(Phi(X)))	42 configs (q85|420: 29, q85|444: 29, q92|420: 29, q92|444: 29, q95|420: 29, q95|444: 29)	0.0000	0.1899	0/11661	0.0257%	1794
3	F_float	CELL, eta = 0.05, Phi; tables ≤ Q_pol	6b/float, turbo210/float	0.0346	0.0758	0/3588	0.0835%	0
5	Phi_go	CELL, eta = 0.25, Phi_go (4:2:0)	go	0.0905	0.0953	0/897	0.333%	0
7	XP	re-encode with the pinned entry, every coefficient	mozjpeg403/islow/default	0.0000	90.0882	0/1794	0.167%	1794
–	ifast	non-member (integer fast DCT)	–	–	36.0506	0/0	–	0
–	islow_dct16	non-member (IJG ≥ 7 default 4:2:0 (DCT-scaled chroma))	–	–	29.5570	0/0	–	0
–	float_dct16	non-member (IJG ≥ 7 float default 4:2:0)	–	–	29.5570	0/0	–	0

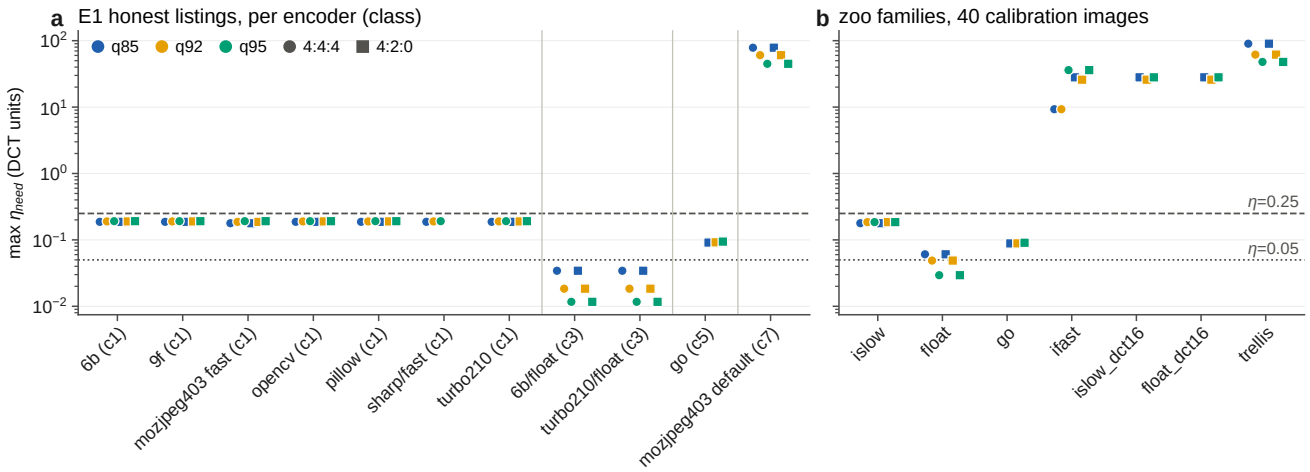


Fig. R1. E1: honest deviation per encoder and class against the cell half-width. (a) largest η_{need} per encoder, quality and sampling over the held-out listings; (b) encoder families on 40 calibration images, including the non-members `ifast`, DCT-scaled chroma and trellis.

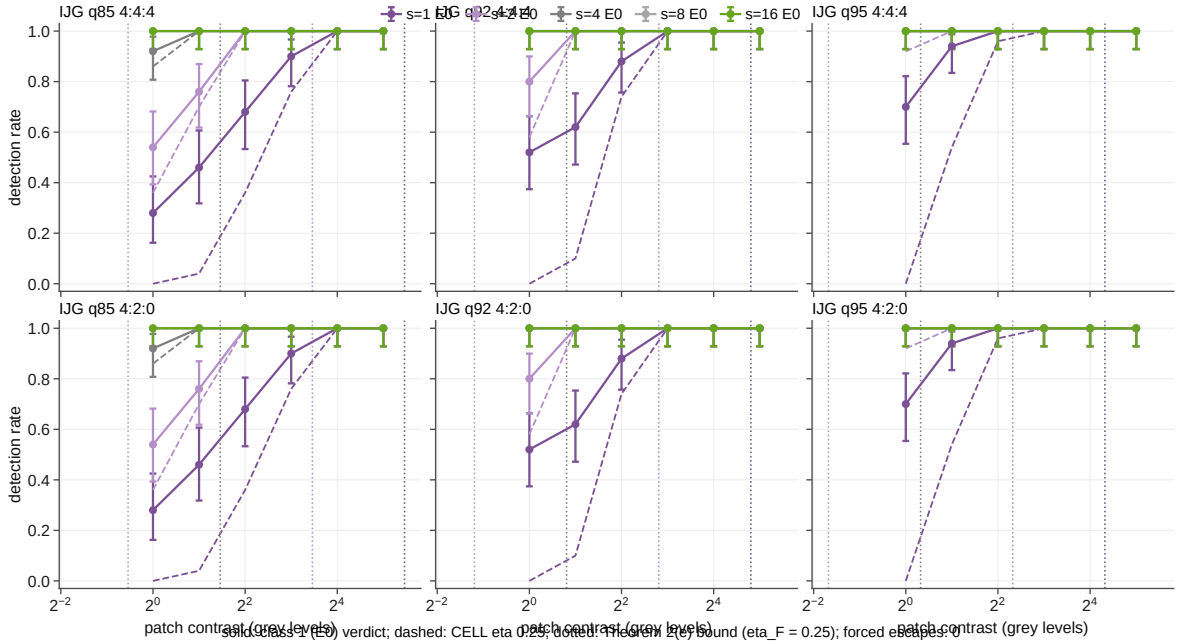


Fig. R2. E2: detection of an $s \times s$ constant patch against its contrast, per upload profile (9,000 listings). Solid: `X0`; dashed: the tolerant cell at $\eta = 0.25$; dotted: the Theorem 2(e) forced-rejection bound. No forced escape was observed.

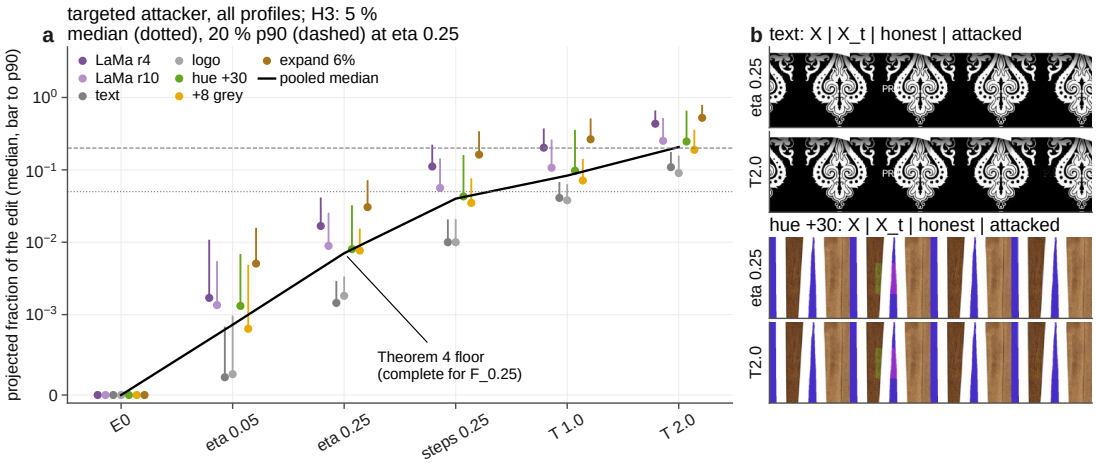


Fig. R3. E3: projected fraction of an edit that reaches the screen under an optimal attacker, from X0 (zero by construction) through the tolerant cells to fixed pixel thresholds; the Theorem 3 floor for any verifier complete for $\mathcal{F}_{0.25}$.

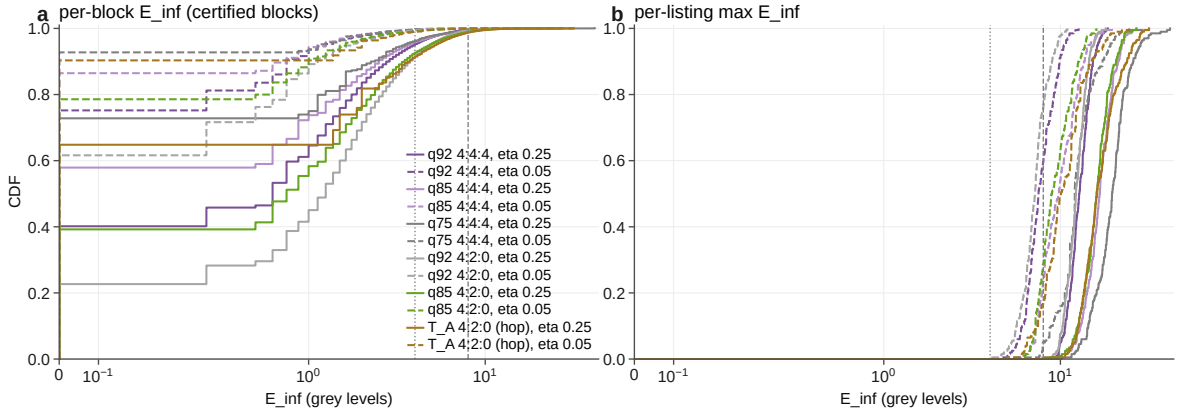


Fig. R4. E3: per-block and per-listing E_∞ distributions for the tolerant classes.

R2 Completeness, soundness and adaptive attackers (E1–E3)

Further E2/E3 numbers. The six E2 upload profiles are q85, q92 and q95 at 4:4:4 and 4:2:0. In E3 the tolerant cell is below the former pixel threshold $T = 2.0$ in every class (one-sided Wilcoxon, Holm-adjusted $p < 10^{-190}$), and a table-grinding attacker realises a median 0.45% (90th percentile 2.3%, $n = 97$) of the edit at $\eta = 0.25$.

R3 Baselines (E4)

Table R4 is the full E4 table (both profiles, every baseline and Δ, the A10 result); the main text prints a condensed version. Table R5 gives the per-pixel tolerant comparator E19 on the same items; its protocol and the X1-cell comparator E21 are described in §R4.

Published-numbers rows. EditGuard and OmniGuard were not run because their repositories grant no licence; CAT-Net [1] passed its licence gate but its weights could not be downloaded (every download from the official folder was refused). These methods are listed in the note of Table R4 instead of as rows; the published numbers we rely on are those quoted in Sec. II (e.g., an off-the-shelf CAT-Net F1 of 0.02 on fully regenerated inpainting [2]).

Further E4 numbers. Null edits (no pixel changed) were excluded and reported (41 listings), and 72 attacked items were skipped because no textured point lay inside the silhouette and the certified region. TrustMark's clean decode on the honest test listings is 298/299 (U85), 297/299 (AMZ) and 299/299 (P-SHOP); after the frozen removal attacker it is 201/299, 215/299 and 209/299. Mesorch at FPR 5% (threshold 0.9999982, honest FRR 7.4%) detects A1 4.7%, A2 6.1%, A3 2.0%, A4 7.1%, A5 6.1%, A6 7.1%, A3sd 6.0%, A6sd 4.0% and A7sd 4.0%. TruFor ran on 2,696 images with a measured peak of 4,361 MB; its honest FRR at AMZ is 3.0% at the FPR-1% threshold (a calibration/test shift). The 150 diffusion attacks took a median of 116.5 s each (peak 6,338 MB).

TABLE R4. E4 detection at a calibrated FPR of 1%. TPR (%) per attack class; FRR (%) on the honest test listings. A3sd, A6sd and A7sd have 50 attacks each, so their rates are whole percent.

Method	A1	A2	A3	A4	A5	A6	A3sd	A6sd	A7sd	FRR
Profile U85 (seller upload, q85)										
CUTLINE tier X (X0)	100.0	100.0	100.0	100.0	100.0	100.0	100	100	100	0.0
tier T (4:4:4 table)	86.9	91.9	100.0	68.7	91.9	89.9	–	–	–	1.0
Sun–He–Tian Δ=4	67.4	79.8	100.0	61.2	80.8	73.7	92	0	96	0.0
Sun–He–Tian Δ=8	62.8	73.7	100.0	57.1	73.7	58.6	90	0	96	0.0
Sun–He–Tian Δ=16	50.0	62.6	99.0	51.0	48.5	42.4	74	0	92	0.0
Signed thumbnail	61.6	73.7	100.0	60.2	78.8	67.7	94	100	96	0.0
pHash	0.0	1.0	2.0	15.3	0.0	0.0	0	38	2	0.3
PDQ	0.0	2.0	13.1	24.5	0.0	0.0	2	78	10	0.0
Birthmark replay	7.0	3.0	23.2	32.7	2.0	14.1	14	90	40	0.0
PROVES-style	0.0	0.0	1.0	6.1	0.0	0.0	0	0	0	0.0
TruFor	0.0	0.0	0.0	0.0	0.0	0.0	0	0	0	0.3
Mesorch	0.0	0.0	0.0	0.0	0.0	0.0	0	0	0	0.0
TrustMark	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0.0
Profile AMZ (after the P-AMZ re-encode)										
CUTLINE tier X (X0)	100.0	100.0	100.0	100.0	100.0	100.0	100	100	100	0.0
Sun–He–Tian Δ=4	28.7	59.0	97.0	55.6	34.0	36.0	90	0	96	0.0
Sun–He–Tian Δ=8	29.9	63.0	97.0	56.6	33.0	36.0	94	0	96	0.0
Sun–He–Tian Δ=16	23.0	52.0	95.0	51.5	22.0	24.0	88	0	96	0.0
Signed thumbnail	18.4	39.0	90.0	55.6	20.0	16.0	82	100	92	0.3
pHash	0.0	0.0	1.0	15.2	0.0	0.0	0	42	4	0.0
PDQ	0.0	0.0	1.0	28.3	0.0	1.0	0	80	10	0.0
Birthmark replay	0.0	0.0	2.0	37.4	0.0	2.0	8	70	42	0.0
PROVES-style	0.0	0.0	0.0	7.1	0.0	0.0	0	0	0	0.0
TruFor	4.6	4.0	4.0	2.0	4.0	4.0	0	0	0	3.0
TrustMark	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0^b	0.0

^{b} by design (TrustMark is a soft binding and does not localise edits). A10 is scored for tier T only: median projected fraction 0.80 under the tier-T table (X0: 0). No baseline dominates CUTLINE in any cell (exact McNemar, Holm $p < 0.05$). Not shown: CAT-Net, EditGuard and OmniGuard (gate failed: published numbers only, see §R3); Mesorch at AMZ (run at U85 only); tier T at AMZ (tier T is unlinked through platform hops in the evaluated design). A6sd: tier-test verdict; the pixels added outside the drawn silhouette are reported separately and never scored as detection.

TABLE R5. E19: per-pixel tolerant comparator. TPR (%) at FPR 1% of the per-pixel tolerant scores on the E4 test items, with the calibrated threshold θ (grey levels) and the honest test false positives (FP, out of 299). CUTLINE tier X is 100.0 on every cell with 0/299 honest. Bold: not significantly below CUTLINE (exact McNemar $p \geq 0.05$).

Score	θ	A1	A2	A3	A4	A5	A6	A3sd	A6sd	A7sd	FP
Profile U85											
`pix_max`	91	27.9	42.4	89.9	11.2	0.0	4.0	30	**96**	54	0
`pix_max_sil`	91	27.9	42.4	89.9	11.2	0.0	4.0	30	8	54	0
`blk_mean`	9.51	52.3	67.7	**100.0**	45.9	6.1	53.5	70	**100**	82	1
`blk_mean_sil`	9.28	54.7	67.7	**100.0**	45.9	7.1	53.5	74	4	86	1
Profile AMZ											
`pix_max`	148	3.4	7.0	7.0	7.1	0.0	0.0	24	62	36	0
`pix_max_sil`	139	3.4	9.0	11.0	9.1	0.0	0.0	24	0	40	0
`blk_mean`	18.15	11.5	37.0	**100.0**	32.3	0.0	8.0	58	**100**	76	3
`blk_mean_sil`	18.15	11.5	37.0	**100.0**	32.3	0.0	8.0	58	2	76	0

TABLE R6. A10: threshold-hugging edits. Median (90th percentile) projected fraction of a targeted edit that passes tier T (table f1 4:4:4), against the X1 cell at $\eta = 0.25$ (E3, q85 4:4:4); X0 passes 0. $n = 100$ per edit.

Edit	tier T median (p90)	accepted	X1, $\eta = 0.25$ (median)
brighten 8, 32 px	0.308 (0.427)	100/100	0.007
expand 6%, $R = 40$	0.796 (0.943)	100/100	0.031
LaMa removal $r = 10$	0.413 (0.762)	100/100	0.009
LaMa removal $r = 4$	0.771 (0.900)	100/100	0.017
logo 24 px, $\alpha = 0.6$	0.146 (0.194)	100/100	0.002
recolour hue 30, 48 px	0.431 (0.724)	98/100	0.007
text 12 px	0.107 (0.218)	100/100	0.001

LaMa denotes the inpainting model of [3].

TABLE R7. Additional experiments: registered predictions and outcomes. Predictions are quoted from the registration without implementation names. E13–E17: registered together; E17b (in the E17 row) and E18–E21: each registered before its own run or analysis; "descriptive": the prediction holds by construction or only describes a rate.

ID	Registered prediction	Outcome
E13 tone	A monotone tone table can hold the product's p95 ΔE_{00} just under the class thresholds while raising the mean / p99 product ΔE_{00} well above a just-noticeable difference (ΔE_{00} > 1–2), at G4 for the minor class and G3 for the moderate class; the cosmetics category cap (p95 ≤ 2.0) limits it to the minor level.	Holds: mean 1.64 (G4), 3.35 (G3); targeted p99 3.16 / 7.03; 100/100 each; cosmetics mean-objective G4 on 88/100 (targeted G3 on 94/100).
E13 *D*	With $k = \lfloor \lvert M_{\text{cert}} \rvert / 51 \rfloor$, a declared-retouch blob or boundary trim reaches G3 (not G2) while removing or replacing a visible product part; the trim changes the visible shape (IoU < 1, Hausdorff > 0) and the removed part is drawn as a hole; $k + 1$ drops to G2.	Holds: blob G3 100/100 (2.95% replaced); trim G3 99/100 (IoU 0.973, Hausdorff 54 px; one G4 from a capture miss); $k + 1$ G2 100/100.
E13 gen.	Generated content placed strictly in uncertified MCUs beyond the certified footprint is never scored by the tier-X test (0 generated pixels in the certified set and in *Sil*); the listing keeps a positive level (G3, background generated-declared) and is not detected. Reported as a residual, not a defect.	Descriptive (true by construction): 0 generated pixels in M_{cert} and *Sil*; G3 30/30 (also with a declared bridge D), i.e., not detected. Adversarial-scene arm not run.
E14	The median miss is 0 in every stratum; a minority have a material capture miss (> 2%), with a per-stratum CP upper bound that we report.	Descriptive: median 0; > 2% on 20, 8, 4 of 100 (bounds 0.277, 0.140, 0.089).
E15	Honest star failures = 0 on every profile; the WF3 star ladders are at the source level on 100% of in-scope honest renditions; every attack case matches its expected (linked, error, level).	Holds: 0 failures (3,603 renditions); 185/185; 38/38. All masters ABO.
E17	The 12 rows reproduce on Besu up to calldata-byte and ECDSA-v differences; if Besu cannot run them, they are marked EDR-only with the reason.	Partly: 11/12; the 65,535-MCU proof is EDR-only (transaction size). E17b: mined with the pool limit raised (−60 gas).
E18	AS9 is supported in its restricted form iff every decoder of the family (libjpeg-turbo and IJG 6b at `islow` with fancy upsampling, Pillow and OpenCV as built) displays Dec* bit-exactly on the certified region of every sampled file; Part A (a re-check of the hashes that E9 recorded) passes iff every recorded hash equals the Dec* hash.	Holds: 3,600/3,600 pairs on 400 files; Part A 202/202 for Pillow, OpenCV and Chrome. Outside the family: §R8.
E19	Where `pix_max` TPR is below CUTLINE with McNemar $p < 0.05$, the paper may say that a per-pixel tolerant test on the same openings misses edits the exact test catches; elsewhere it must say tolerance does not cost detection.	Below on 17/18 cells (U85 0.0–96.0%, AMZ 0.0–62.0%; unchanged under Holm's correction); not on A6sd at U85 (96.0%, $p = 0.5$).
E20	The G2 renditions at 500 × 375 are mainly wide or tall products iff ≥ 80% of the G2 renditions at 500 × 375 lie outside an aspect band around 4:3 and the G2 share inside the band is materially lower; otherwise the dominant recorded cause replaces it.	Met literally (92.2% outside; 58.8% vs 100%), wrong in substance: every G2 is *partial* from the 4:3 crop, square canvases included (26/26).
E21	Where the calibrated X1-cell TPR is below CUTLINE with McNemar $p < 0.05$, the paper may say that even the cell test on the same openings misses edits; elsewhere it must say that a per-cell tolerant test detects about as much and that the advantage is the zero excess and the one-point acceptance set.	Below on 0/18 cells: 100% everywhere ($\eta = 0.178, 0.181$), honest 1/299, 0/299; fixed $\eta = 0.25$: 100%, 0/299.

Tier-T test FRR per profile. f0 4:2:0, f0 4:4:4, f1 4:2:0, f2, f3 and f4 (both samplings): 0/299 each; f1 4:4:4: 3/299 = 1.003% (95% CP interval [0.0021, 0.0290]). The order-statistic guarantee is $P(\text{FRR} > 1\%) \leq 0.0495$ per profile, so one profile in ten above 1% is within its reach, but the registered point prediction fails for that profile.

A6sd and the honest coverage check. The A6sd tier-test column was predicted to stay below 100% (the ring touches only boundary blocks), but CUTLINE detected 50/50 on both profiles; the likely mechanism, not diagnosed per listing, is that the ring meets certified MCUs because A is dilated beyond *Sil*. The registered honest tier T coverage check (every honest 256-px listing tests at least one block) fails as written on 1/299: every existing listing tests at least one block (minima 618 and 594 on the two kernels), but one thin product has no 256-px listing at all (refused below the upload floor). The rule that a claim certifying nothing is G0 is unchanged.

R4 Residual surface and tolerant comparators (E13, E19, E21)

The additional experiments were registered after the main results (E1–E12), each before its own run or analysis (registration record: §R1.3). E13–E17 were registered together, with their predictions, before any run on test masters; E17b and E18–E21 were registered individually afterwards. Table R7 quotes every registered prediction with its outcome; the E19 and E21 predictions quote the registration's own rule for what the paper may claim. E13 measures how far a white-box seller who stays inside the admissible programs can change the displayed product while keeping G3 or G4, starting from the honest G4 constant-white listing of each E5 test master (Fig. R5). In its tone row, the cosmetics category cap makes a tone table whose product p95 ΔE_{00} exceeds 2.0 major (G2) for cosmetics and apparel listings, so only the minor class remains. The E19 table is Table R5 (§R3). **Tone (prediction holds).** A monotone 3 × 256 table searched over five families and 12 directions keeps the tone class while moving the product: at G4 (minor) the median product mean ΔE_{00} is 1.64 (90th percentile 1.86) with the mean objective, and the targeted 99th-percentile ΔE_{00} reaches 3.16; at G3 (moderate) the mean is 3.35 (p90 3.83) and the targeted p99 7.03 (maximum 9.60); 100/100 listings stay at their level in each case. Under the cosmetics cap the mean-objective table stays minor (G4) on 88/100, and the targeted table reaches G3 on 94/100 with p99 5.25. A G4 label prints no tone disclosure; G3 prints "colour adjusted" with the ΔE_{00} statistics. The search gives lower bounds on a white-box optimum; the class bounds any table at $\Delta E_{00} \leq 4$ (G4) and ≤ 10 (G3) on certified product pixels.

Declared retouch (prediction holds). With $k = \lfloor \lvert M_{\text{cert}} \rvert / 51 \rfloor$ declared MCUs (median $k = 61$) a blob recolour or inpaint stays G3 on 100/100 while replacing a median 2.95% of the product pixels; a boundary trim stays G3 on 99/100

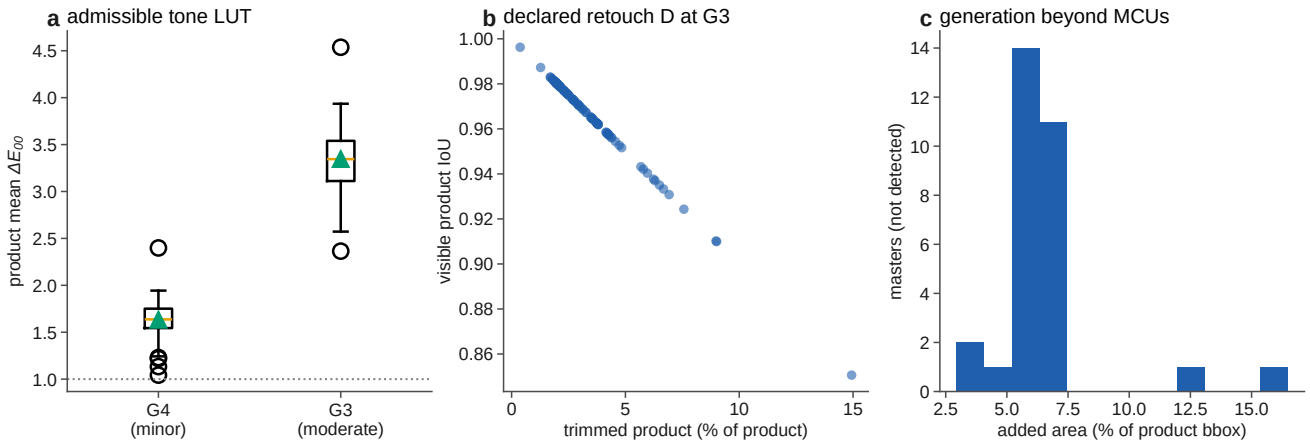


Fig. R5. Residual surface of E13. (a) Product mean ΔE_{00} reached by the mean-objective monotone tone table while the tone class stays minor (G4) or moderate (G3); triangle: mean; dotted line: $\Delta E_{00} = 1$. (b) Declared retouch D with $k = \lfloor |M_{\text{cert}}|/51 \rfloor$ MCUs used to trim the product boundary: visible product IoU against the trimmed fraction (99 listings at G3). (c) Diffusion content added wholly in uncertified MCUs beyond the certified footprint (30 listings, all G3, none detected): added area relative to the product's bounding box.

with visible-product IoU 0.973 and Hausdorff distance 54 px (medians); with $k + 1$ MCUs all 100 drop to G2. The printed evidence says "retouched areas marked", draws the D MCUs as holes in *Sil* and prints the retouched fraction. The one trim at G4 is a capture miss: no certified MCU of that master contains ground-truth product pixels, so nothing was declared.

Generation beyond the certified footprint (prediction holds; a residual). Stable Diffusion 1.5 inpainting of a region in uncertified MCUs beyond the footprint (30 masters) put 0 generated pixels into M_{cert} or *Sil*; every listing stayed G3 with "background AI-generated" (median gap to the product 13.4 px, about the dilation), also with a declared bridge D (median 64 MCUs; G3 30/30). Such content is not detected: a buyer can only see that it lies outside the drawn outline. The E4 A6sd ring, which meets certified MCUs, was detected 50/50.

Not run. Adversarial scene content against the capture segmenter (the fourth E13 arm) was removed from scope during development; capture coverage loss is measured descriptively by E14 (§R5).

Reference-holding rows of E4 (post-registration analysis, disclosed). After the E4 results were known, we took its reference-holding tolerant tests, which rebuild the same declared-program reference from the openings and are calibrated at matched FRR, as a first equal-information comparator; no new run. This is an extraction made with the values known, not a rule fixed before any result, and these coarse features are not the strongest tolerant test on the same openings, so the manuscript uses E19 instead. At FPR 1% on U85, Sun–He–Tian detects 67.4–100 ($\Delta = 4$), 57.1–100 ($\Delta = 8$) and 42.4–99.0% ($\Delta = 16$) per class except A6sd (0%), and the thumbnail 60.2–100%; at AMZ the tolerant tests fall to 16.0–97.0% (A6sd and A7sd aside); CUTLINE detects 100% everywhere with 0/299 honest convictions per profile (Table R4).

E19: global-threshold per-pixel comparator. Its registration (committed in afd30d3 at 2026-10-07T23:48:06Z before the run, with no score computed) fixes the scores, the calibration and the interpretation rule. The runner reuses the frozen E4 items and data factory unchanged (masters, splits, attack draws, diffusion attacks, null-edit rule). For each listing, $Z = \text{Dec}^*$ of the displayed bytes (the U85 upload, or the P-AMZ L1000 rendition for AMZ), R the reference CUTLINE rebuilds from the openings (pushed through the declared hop with its exact kernel for AMZ) and C the certified pixels (pushed through the hop for AMZ). The primary score `pix_max` is $\max_{p \in C,\, c} |Z - R|$, a pixel test with one global threshold; `pix_max_sil` restricts it to $C \cap \mathit{Sil}$; `blk_mean` is the largest mean $|Z - R|$ over the certified pixels of an 8×8 block of the displayed grid (blocks with ≥ 32 certified pixels); `blk_mean_sil` is its *Sil* version. The FPR-1% threshold is the maximum over the 299 honest calibration listings per profile; a test item is flagged iff its score exceeds it. CUTLINE is the frozen tier-X verdict of the E4 run on the same item keys; the paired test is an exact two-sided McNemar test. 2,696 items were scored (0 errors); 6 were skipped (no listing or no textured point) and 28 null edits excluded, as in E4. Table R5 gives the result. By the registered rule, `pix_max` is significantly below CUTLINE on 17 of 18 (class, profile) cells; Holm's correction over the 18 cells changes no decision (the largest significant p, 0.002, is below its Holm threshold 0.025). On A6sd at U85 (48/50, $p = 0.5$) the difference is not significant. The best variant, `blk_mean`, ties CUTLINE ($p \geq 0.05$) only on A3 and A6sd on both profiles, at an honest test FPR of 1/299 (U85) and 3/299 (AMZ). The *Sil* variants lose A6sd because the ring lies outside *Sil*.

E21: the X1 cell as an equal-information comparator. Registered in 076b16d (2026-10-08T00:55:43Z) before the run, on the same 2,696 E4 items. The score is the smallest slack that accepts every tested coefficient, $\eta_{\text{need}} = \max_k(|c_k Q_k - F_k| - Q_k/2)$ in DCT units, where F is the DCT of the reference front end, over the blocks of M_{cert} at U85 and, at AMZ, over the rendition MCUs that contain a pushed certified pixel, against the hop reference that a verifier rebuilds from the openings

(Dec* of the exact upload, resized by the declared hop). Thresholds are the calibration maxima (FPR 1%): $\eta = 0.178$ (U85) and 0.181 (AMZ). The cell test detects 100% of every class on both profiles (McNemar $p = 1$ against CUTLINE on all 18 cells; against `pix_max` $p \leq 0.002$ on 17 cells, $p = 0.5$ on A6sd at U85) with 1/299 (U85) and 0/299 (AMZ) honest test alarms. Every attacked listing needs $\eta_{\text{need}} \geq 1.0$ (U85; 1.026 at AMZ), every honest test listing at most 0.183, so the result does not depend on where in [0.18, 1) the threshold lies; the design's fixed $\eta = 0.25$ also detects 100% with 0/299 honest alarms, and $\eta = 0.05$ rejects every honest listing. Against the reference of E19 at AMZ (the original pushed through the hop, without the upload's quantisation) the cell test is weaker (48–100%; descriptive), so the transcoding-aware reference matters. By the registered rule, a well-designed per-cell tolerant test detects as much as CUTLINE on these attacks; CUTLINE's advantage over it is the zero adversarial excess (E3) and a one-point acceptance set with no slack parameter (the fixed $\eta = 0.25$ is also uncalibrated, but admits a whole cell), not the detection rate.

R5 Region coverage (E5, E14)

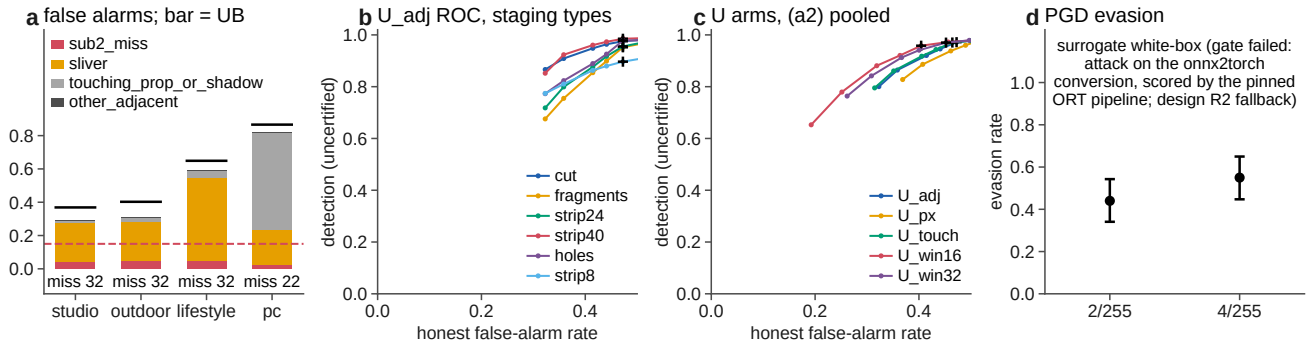


Fig. R6. E5 region coverage (H5; main Sec. VIII-C). (a) False alarms of $U > \tau_U$ per honest workflow type, split by mechanical cause (sub-2% capture miss, sliver, touching prop or shadow, other adjacent); the black bar is the one-sided upper bound used by (a1), the dashed line the 15% criterion. (b) ROC of U per mask-only staging type (detection among uncertified staged listings against the honest false-alarm rate, $\tau \in \{8, \ldots, 128\}$; + marks $\tau_U = 32$). (c) The same for the registered U variants, pooled over the (a2) types. (d) Surrogate white-box PGD evasion at $\epsilon = 2/255$ and $4/255$ with 95% CP intervals.

TABLE R8. E5 per stratum. k/n per honest workflow type (false alarms of $U > \tau_U$, with the one-sided 95% upper bound) and per mask-only staging type (detections, with the lower bound); last row: captures re-drawn by the ground-truth reshoot rule.

Measure	White	Scene	DIS
Studio, false alarms (UB)	16/80 (0.288)	33/92 (0.449)	28/96 (0.377)
Outdoor, false alarms (UB)	17/80 (0.302)	31/92 (0.427)	35/96 (0.453)
Lifestyle, false alarms (UB)	39/80 (0.585)	53/92 (0.663)	66/96 (0.765)
Product-conditioned (pc), false alarms (UB)	47/51 (0.973)	51/64 (0.875)	48/63 (0.847)
Cut, detected (LB)	1090/1096 (0.989)	996/1014 (0.974)	985/1038 (0.936)
Fragments, detected (LB)	394/398 (0.977)	455/473 (0.944)	724/781 (0.910)
24-px strips, detected (LB)	365/368 (0.979)	600/626 (0.943)	733/783 (0.920)
40-px strips, detected (LB)	1099/1101 (0.994)	1082/1097 (0.979)	1060/1092 (0.961)
Holes, detected (LB)	521/528 (0.975)	279/290 (0.938)	135/138 (0.945)
Ground-truth reshoots	20/100	8/100	4/100

Reported types. Detached completion 95/100 (CP lower bound 0.898), detached twin 33/77 (0.333), low contrast 43/49 (0.773), occluder 47/50 (0.852), prop 48/48 (0.939), 8-px strips 373/416 (0.869). At $\tau_U = 32$ the registered U variants detect / falsely alarm: U_{adj} 10,891/11,239 and 478/1,298; U_{px} 11,003 and 525; U_{touch} 10,876 and 470; U_{win16} 10,734 and 411; U_{win32} 10,869 and 458. Design effects (Kish): studio $\hat{\rho} = 0.273$, deff 2.56; outdoor 0.381, 3.17; lifestyle 0.109, 1.38. The mask-reuse check flipped 2/99 U decisions (contingency at 5% not triggered), and two independent runs produced byte-equal public masks on 50/50 listings. Under the surrogate PGD at $\epsilon = 4/255$, evasion per staging type is cut 15/28, fragments 6/14, holes 6/9, 24-px strips 10/16, 40-px strips 17/30 and 8-px strips 1/3. The evaluation's ground-truth reshoot rule (more than 2% of the product missed) fired with U²-Netp [4] as capture model on 13/17 DIS capture attempts (miss median 0.287) and with BiRefNet-lite on 4/117 (DIS), 9/119 (scene) and 21/114 (white) capture attempts, i.e. 4, 8 and 20 of the 100 accepted masters per stratum (Table R8 counts per accepted master). The (a1) denominators exclude the listings of these 32 masters

(pasted types) and 22 incomplete pc listings. BiRefNet-lite as the public (listing-side) segmenter does not help: 195/268 false alarms.

Hand audits (protocol). Two registered audits are reported as agreement only and change no mechanical label (results: §R6). Audit 1 covers the 30 honest false-alarm listings that come first by a fixed hash of the listing key (domain string `CUTLINE-E5-causeaudit-v1`; `cause_audit_keys` in `experiments/s9/e5/e5.json`), as 145 units (97 numbered components and 48 residual groups). Audit 2 covers the 30 masters of the ground-truth contact sheet (`experiments/s9/e5m/audit_sheet.jpg`). Before rendering, the components and causes were recomputed with the unchanged code path and checked against `e5_honest.jsonl`. The rendered items, the unit definitions and the answer categories are in `experiments/s9/hand_audit/` (`manifest.json`, `README.md`, regenerated by `build_hand_audit.py`). Five annotators labelled each audit jointly on the labelling page (`audit_page.html`) and agreed on one consensus label per unit or image. The mechanical label or the rule's verdict was shown beside the choices. The labels were exported on 2026-10-11 to `labels_audit1.json` and `labels_audit2.json`. They are scored by `score_hand_audit.py` (stdlib), whose rule was committed in b660ce0 before any label existed, into `hand_audit_result.json`, which `experiments/s9/s9_exit.py` reads for rows 3 and 5 of the S9 exit report. Because the labels are consensus labels, no inter-annotator agreement is reported.

E14: coverage against ground truth. For the BiRefNet-lite capture of each E5 test master (100 per stratum), the fraction of ground-truth product pixels outside *Sil* equals that outside *A* for every master (the whole-frame listing certifies the placed region exactly). Its median is 0 in every stratum; its mean is 4.7% (white; 95% CI 2.2–7.7%), 4.5% (scene; 1.4–8.4%) and 0.39% (DIS; 0.14–0.75%), 3.2% over all 300. More than 2% is missed on 20/100 (white, CP upper bound 0.277), 8/100 (scene, 0.140) and 4/100 (DIS, 0.089), 32/300 overall (0.141): exactly the masters that the evaluation's ground-truth reshoot rule re-drew. A deployed app has no ground truth, so these misses remain an operational residual. The registered prediction (median 0, a minority with a material miss) holds; this is the first ground-truth measurement of how often AS7 fails.

TABLE R9. Workflow families WF1–WF3. Share of honest listings at the target level, *U* printed only (the setting in force after gate E5 failed) and *U* as a level criterion (as first computed); 95% CP intervals. WF1 and WF2 pass; WF3 is deployable on one-hop (star) ladders only: every cascade row straddles 0.5 and the E6 P-SHOP cascade rows are below it (bounded 0.481; off/on 50/103 = 0.485, [0.386, 0.586]).

Family	fill	capture	target	*U* printed only	*U* a criterion
WF1 constant	bounded	BiRefNet-lite	G4	300/300 [0.988, 1.000]	288/300
WF1 constant	bounded	U²-Netp	G4	50/50 [0.929, 1.000]	50/50
WF1 constant	off / on	BiRefNet-lite	G4	100/100, 100/100	96/100, 99/100
WF1 constant (halo-free variant)	bounded	BiRefNet-lite	G4	50/50	46/50
WF2 lifestyle+pc	bounded	BiRefNet-lite	G3	500/500 [0.993, 1.000]	150/500 [0.260, 0.342]
WF2 lifestyle+pc	bounded	U²-Netp	G3	89/89	25/89
WF2 lifestyle+pc	off / on	BiRefNet-lite	G3	168/168, 168/168	46/168, 49/168
WF2 lifestyle+pc (halo-free variant)	bounded	BiRefNet-lite	G3	89/89	23/89

WF3 (share at the source listing's level; WF2 and halo-free sources through P-AMZ / P-SHOP), *U* printed only:
star 440/440, 447/447; cascade 150/290 = 0.517 [0.458, 0.576], 150/297 = 0.505 [0.447, 0.563]
U a criterion (as first reported): star 440/440, 447/447; cascade 250/290, 255/297
WF3 of WF1 (E6 ladders, bounded fill), *U* printed only: star 895/895, 918/918;
cascade P-AMZ 299/596 = 0.502 [0.461, 0.543], P-SHOP 298/619 = **0.481** [0.441, 0.522]

R6 E5 hand audits

The E5 registration includes two hand audits, reported as agreement only; neither changes a mechanical label or a decision. Audit 1 covers the 30 honest false-alarm listings that come first in the order of $H(d \,\|\, key)$ with domain string d = `CUTLINE-E5-causeaudit-v1`. Audit 2 covers the 30 accepted masters of the ground-truth contact sheet (15 white, 15 scene, drawn by $H(d' \,\|\, id)$ with d' = `CUTLINE-E5-audit-v1`). The scoring rule was committed before any label existed (commit b660ce0, 2026-10-07 20:45:36 UTC). *Protocol.* Five annotators labelled each audit jointly by eye, from rendered overlays and zoom crops and without re-running any rule, and reached one consensus label per unit (audit 1) or image (audit 2); the labels are therefore not independent, and no inter-annotator agreement is reported. The mechanical label (audit 1) or the rule's verdict (audit 2) was shown beside the answer choices, so the audits are not blind. *Units (audit 1).* The 30 listings have 3,410 C_{adj} components, most of 1–3 px. Each listing therefore shows its largest components until they hold ≥ 90% of the C_{adj} pixels (at most 8), plus the largest component of each mechanical cause not yet shown. All other components form one residual group per mechanical cause. This gives 145 units (97 components, 48 groups). Each unit receives one of the four causes or "unclear". *Statistics.* The registered unit is the listing for audit 1 and the image for audit 2. A listing agrees iff every one of

TABLE R10. E5 hand audits (consensus labels of five annotators). Top: agreement statistics. Bottom: audit 1 confusion matrix, mechanical cause (rows) × consensus label (columns); no unit was mechanically a sub-2% miss.

Statistic	k/n	rate	95% CP
Audit 1: false-alarm causes (30 listings, 145 units)			
Listing agreement (primary)	5/30	16.7%	5.6–34.7%
Label-set agreement	22/30	73.3%	54.1–87.7%
Unit agreement	95/145	65.5%	57.2–73.2%
Unit agreement, "unclear" excluded	95/113	84.1%	76.0–90.3%
Pixel-weighted unit agreement	689,113/765,008 px	90.1%	–
Cohen's κ (units)	–	0.49	–
Audit 2: ground-truth validity (30 masters; the rule accepted all 30)			
Judged valid (primary)	30/30	100%	88.4–100%
white / scene	15/15, 15/15	100%	78.2–100%

Mechanical ↓ / human →	miss	sliver	touch.	other	unclear	total
sliver	0	35	0	0	9	44
touching prop / shadow	3	6	52	7	19	87
other adjacent	0	1	1	8	4	14
total	3	42	53	15	32	145

its units agrees, and "unclear" counts as a disagreement. Intervals are two-sided 95% Clopper–Pearson. Table R10 gives the results.

The listing statistic (5/30) is strict: most listings have several units, and one disagreeing or "unclear" unit makes the listing disagree. The set of causes agrees in 22 listings, and the pixel-weighted agreement (90.1%) exceeds the unweighted one (65.5%), so the disagreements lie mostly in small components; 32 of the 50 disagreeing units are "unclear". Three units that the rule calls touching props or shadows were read as product misses, consistent with the capture misses measured in E14. The audit supports the cause shares, approximately so for small regions, and does not change the mechanical counts. Audit 2 agrees with the frozen validity rule on every image (pilot reference: 38/40).

R7 Verified hops (E6, E15, E20)

E15: confirmatory rerun. The post-fix code was rerun as a pre-registered confirmatory test on 50 masters never used by any detection or level test (drawn by hash from the manifest ranks beyond the E1–E6 sets, which E12 had only timed; all 50 are ABO, so Objectron and CO3D are not covered): 150 uploads, 1,460 items, 3,603 honest renditions. Gate E6 passes: 0 honest star failures on P-AMZ (n = 678), P-SHOP (687), P-GMC (273), P-GMC/go (123), P-SOC (100) and P-PIN (50); 38/38 attack cases as expected; joint aspect maximum 0.0136 ($\leq$ 0.02); Go class-5 renditions linked at every size (1000 × 750: 32/32); WF3 star renditions at the source level 185/185 (P-AMZ). Cascades keep the source level for 0.37–0.39 (reported, no criterion).

E20: why 500×375 renditions end at G2. Registered in afd30d3 (2026-10-07T23:48:06Z) before the analysis, with the hypothesis that the cause is wide or tall products; the analysis reads only stored records. The N sizes (500 × 375, 999 × 749, 1000×750, 1500×1125) are an E6 test template, not a size of any modelled platform profile (those list only long-side, width and fit sizes): the template takes the largest 4:3 crop of the canvas centred on the certified box and resizes it. A hop is *partial* (G2) when its crop does not contain the parent's MCU-rounded certified box. Every G2 rendition at P-AMZ 500×375 records the reason *partial* (255/269 in the paper's run: 232 alone, 23 with U; 255 of the 257 after the fixes; 48/50 in E15), because the product box fills at least 90% of the canvas, so any canvas whose aspect is not near 4:3 loses certified MCUs (a median 34% of the canvas is cut away for G2 renditions, 2% for the others). Square canvases end G2 on 26/26, tall ones 140/140 and wide ones 89/103; only canvases within about 2% of 4:3 escape (1 of 9 is G2). The registered rule is met literally (92.2% of the G2 renditions lie outside the ±10% band; G2 share 58.8% inside vs 100% outside), but the hypothesis is wrong in substance: the cause is an aspect different from the crop's, not a wide or tall product. Output sizes that are not MCU multiples play no role: of the one-hop aspect-preserving long-side, width and fit renditions with non-aligned outputs, none is partial (3,607 in the paper's run and after the fixes, 1,208 in E15, 546 in W3); their G2 share is that of their source. WF3 claims therefore cover one-hop ladders of aspect-preserving resizes, as in every real profile we model; the WF3 star rows contain no cropping hop. No modelled platform crops, so WF3 on platforms that crop to a fixed aspect (e.g. square thumbnails) is untested.

R8 Displayed pixels under AS9 (E18)

Registration. Registered in afd30d3 (2026-10-07T23:48:06Z) before any measurement. The E9 container differential scored parse agreement only (decode errors, dimensions, component count and, for the libjpeg builds, coefficients); E18 asks which decoders and settings display $\mathrm{Dec}^* = \mathrm{Dec}^*(\mathrm{file})$ (jpeglib libjpeg-turbo 2.1.0, `islow`, fancy upsampling) bit-exactly.

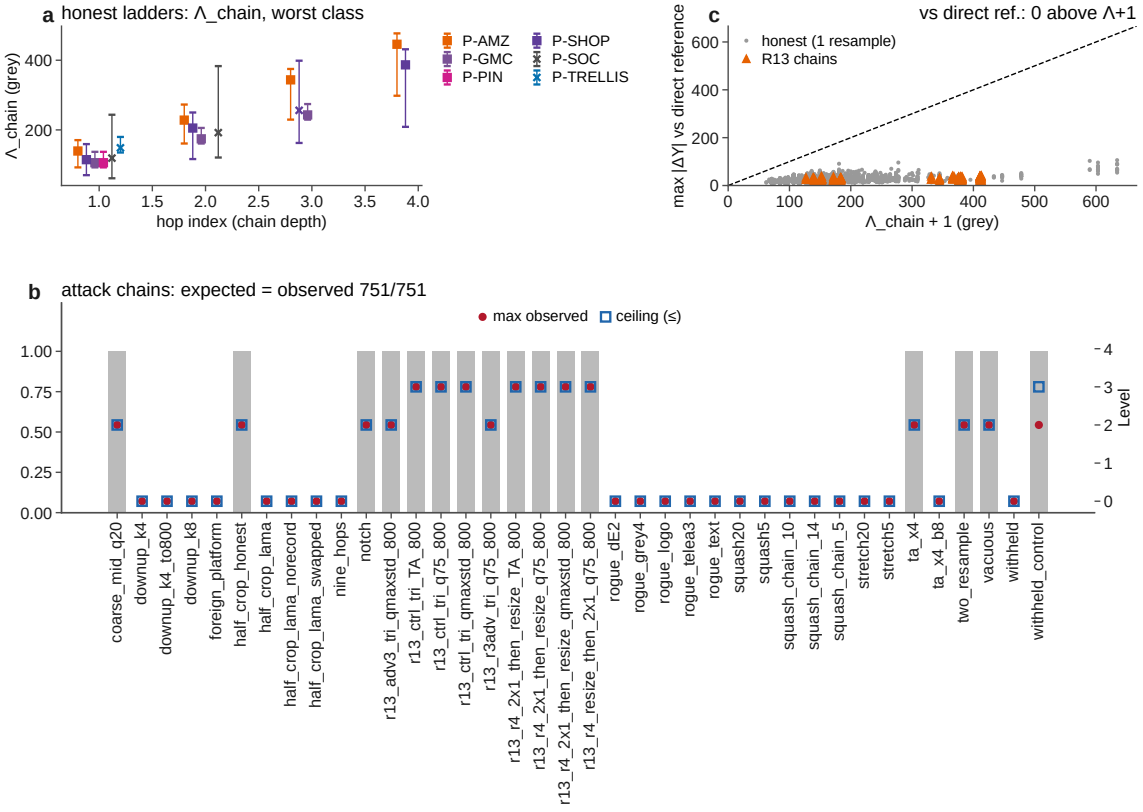


Fig. R7. E6 verified hops after the three disclosed fixes (the pre-registered version is Fig. R8). (a) Λ_{chain} and worst class along honest ladders per platform; (b) attack chains: share linked, maximum observed level and the registered level ceiling, 751/751 renditions as predicted (case labels, e.g. `r13_ctrl_tri_q75_800`, are the artifact's case identifiers: control or adversarial chain, kernel, hop table, width); (c) maximum luma deviation from the direct reference against $\Lambda_{\text{chain}} + 1$ on footprint-certified pixels.

Part A (recorded hashes). The pixel hashes that E9 recorded for the 202 accepted V11 container files (SHA-256 of the RGB bytes for Pillow and OpenCV; FNV-1a-32 of the canvas bytes, alpha skipped, for headless Chrome) equal the same hash of Dec* on 202/202 for each of the three decoders.

Part B (fresh decode). 400 accepted files: the 202 V11 files and, for the first 40 E4 test masters, the U85 and AMZ uploads (4:4:4, 40 each), the P-AMZ L1000 and P-SHOP W1200 renditions (4:2:0, the latter progressive; 40 each) and the P-AMZ 500 × 375 rendition (4:2:0, 38 built); 182 files are 4:4:4 and 218 are 4:2:0. The certified region is CertMask, pushed through the hop for renditions. Decoders: the 24 jpeglib builds (IJG 6b, 7, 8, 8a–8d, 9, 9a–9f; libjpeg-turbo 1.2–2.1; mozjpeg 1.0.1–4.0.3) at `islow` with fancy upsampling, the grid {`islow`, ifast, float} × {fancy, no fancy} for turbo 2.1, 6b and 9f, Pillow 12.3.0 (libjpeg-turbo 3.1.4.1), OpenCV 5.0 (libjpeg-turbo 3.1.2), Go `image/jpeg` and headless Chrome (43 builds and settings), plus our NumPy port of Dec* as a sanity check. The registered criterion: `AS9` holds in its restricted form iff every member of the family is bit-exact on the certified region of every sampled file. Table R12 gives the outcome: the criterion holds, and every exact decoder is also exact on the whole image.

IJG 7 and later upsample chroma by DCT scaling instead of the fancy triangle filter, which is why every 4:2:0 file differs. `AS9` therefore names libjpeg-turbo and IJG 6b at `islow` with fancy upsampling (the library defaults) and Pillow and OpenCV as built; it no longer names "IJG libjpeg" in general.

Run notes (disclosed). (i) jpeglib 1.0.2's `SpatialJPEG.load()` ignores the IDCT and upsampling settings passed to `read_spatial` and uses the library defaults (`islow`, fancy). The first grid pass thus decoded every setting at the defaults; those lines were discarded and the grid re-run with the settings passed to `load()`. The default lines, and Dec* itself (whose settings are the defaults), are unaffected. (ii) The first decode unit was killed by its 2.5-GB memory cap while the NumPy port decoded the 8192 × 4096 V11 file; on the restart the port and Chrome skip files above 4 MP (that one file, recorded as skipped), and the second unit recorded an out-of-memory event only after every stage had written its lines. (iii) No stored tie-flip files exist (E3 keeps E_∞ summaries only), so none are in the sample.

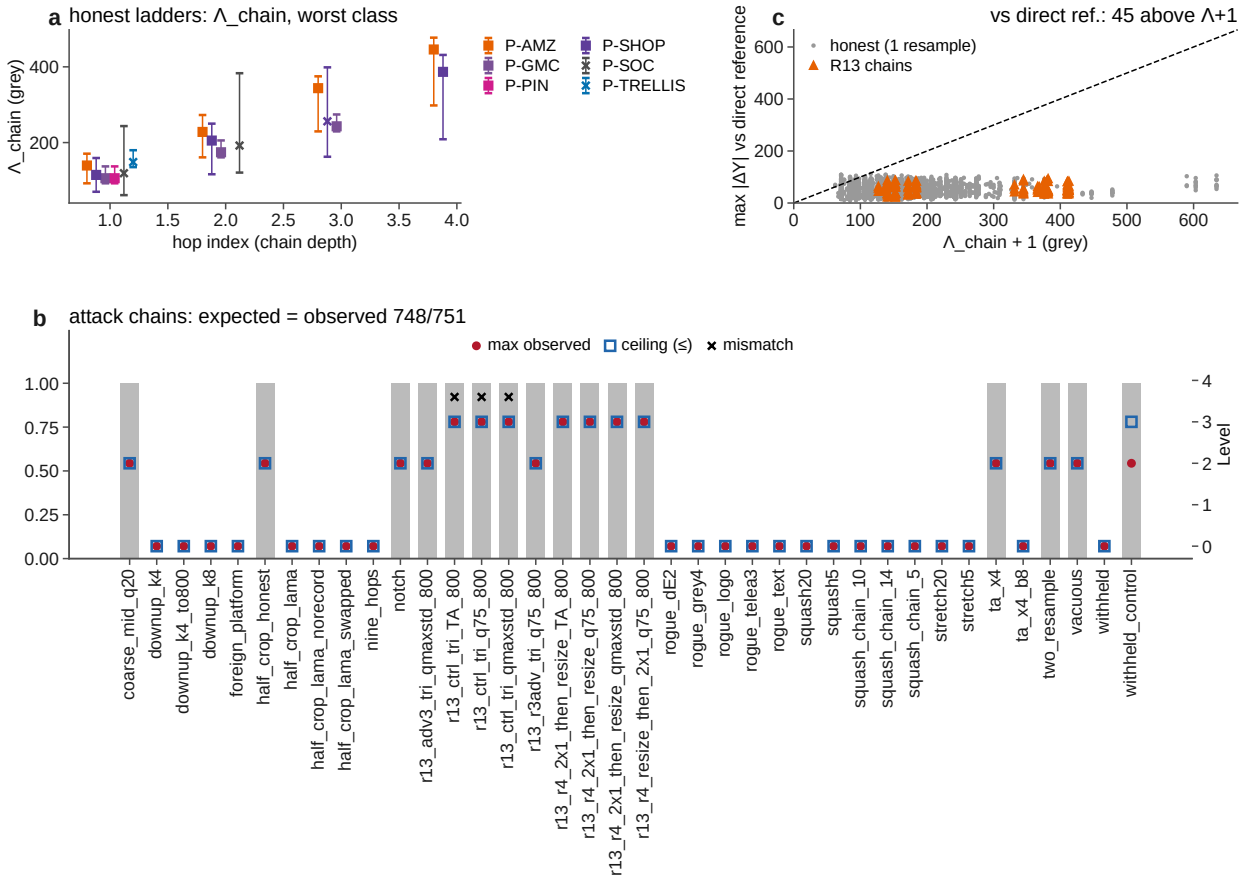


Fig. R8. E6 as pre-registered (before the three disclosed fixes). Panels as in Fig. R7. Panel (c) counts 45 renditions above $\Lambda_{\text{chain}} + 1$: it excludes the one rendition with a rounded non-integer composed crop that the 46/8,603 of the text includes.

R9 Gas costs (E9, E17)

Table R13 lists every gas row.

TABLE R13. CUTLINE V1 GAS PER OPERATION on EDR and on Besu 26.8.1 (receipt gasUsed including the 21,000 base and calldata; floor: EIP-7623), with the mean gasUsed under E10 load (co-located Besu devnet). Besu vs EDR: = equal; cd calldata-byte difference only; floor: floor-bound; v (±26): calldata plus signature-v difference (±1 × 26 gas: recover65 short-circuits, v = 28 costs 26 more); est.: `eth_estimateGas` of a view (client-specific estimator, not a receipt); E17, E17b: court rows measured on Besu in those experiments.

Operation	EDR (gas)	Besu (gas)	Besu vs EDR	Calldata (B)	Floor	E10 mean (gas)
`Adjudicator`						
`deploy`	2,677,250	2,677,250	=	-		-
`finalizeAppeal.cleared`	61,075	61,075	=	36		-
`finalizeChallenge.FAILED`	100,649	100,649	=	36		-
`finalizeChallenge.FAILED_with_2_PASS`	100,649	100,649	=	36		-
`finalizeChallenge.PASSED`	65,111	65,111	=	36		-
`finalizeChallenge.UNDECIDED`	65,149	65,149	=	36		-
`openAppeal`	115,830	115,830	=	36		-
`openChallenge.e420_with_header`	179,419	179,419	=	772		179,897
`openChallenge.tierT`	191,892	191,892	=	548		-
`setClass.executing_vote`	89,020	89,020	=	100		-
`setClass.first_vote`	63,148	63,148	=	100		-
`setClass.vote`	46,152	46,152	=	100		-
`setEPEntry.executing_vote`	111,492	111,492	=	100		-
`setEPEntry.first_vote`	66,258	66,258	=	100		-
`setEPEntry.vote`	49,262	49,262	=	100		-
`setKernelFamily.executing_vote`	90,116	90,116	=	164		-

continued on the next page

Table R13 (continued)

Operation	EDR (gas)	Besu (gas)	Besu vs EDR	Calldata (B)	Floor	E10 mean (gas)
setKernelFamily.first_vote	63,834	63,834	=	164		-
setKernelFamily.vote	46,838	46,838	=	164		-
setTierTable.executing_vote	183,408	183,408	=	132		-
setTierTable.first_vote	68,499	68,499	=	132		-
setTierTable.successor.executing_vote	171,466	171,466	=	132		-
setTierTable.successor.first_vote	68,523	68,523	=	132		-
setTierTable.successor.vote	51,527	51,527	=	132		-
setTierTable.vote	51,503	51,503	=	132		-
submitVerdict.FAIL_first	102,553	102,553	=	100		-
submitVerdict.FAIL_repeated	68,353	68,353	=	100		-
submitVerdict.PASS_first	82,492	82,492	=	100		-
submitVerdict.PASS_next	65,392	65,392	=	100		-
BlockCourt						
deploy	2,787,735	2,787,735	=	-		-
proveBlockFailure.area_boundary	408,832	408,832	E17: =	6,468		-
proveBlockFailure.area_interior	490,283	490,295	E17: cd	10,820		-
proveBlockFailure.boundary_mcu_with_fill	452,620	452,620	E17: =	9,284		-
proveBlockFailure.capture_200mp_l0	4,708,890	4,708,890	E17: =	119,748	yes	-
proveBlockFailure.capture_24mp_l0	1,239,380	1,239,350	E17: cd, floor	32,996	yes	-
proveBlockFailure.deep_tree_wpC	898,320	898,350	E17: cd, floor	28,644	yes	-
proveBlockFailure.design_12mp_nD1024	1,123,310	1,123,310	E17: =	30,980	yes	-
proveBlockFailure.design_12mp_nD256	1,026,770	1,026,800	E17: cd, floor	27,908	yes	-
proveBlockFailure.design_12mp_nD65535	9,256,610	9,256,550[a]	E17b: cd, floor	289,028	yes	-
proveBlockFailure.design_bound_12mp_l0_nD0	994,330	994,330	E17: =	26,884	yes	-
proveBlockFailure.triangle9_probe	741,639	741,639	E17: =	19,140		-
proveBlockFailure.triangle_6cells	627,194	627,170	E17: cd	14,788		-
CaptureAnchor						
anchorBatch.count1	118,983	118,983	=	100		-
anchorBatch.count64	101,883	101,883	=	100		-
anchorBatch.count8	101,883	101,883	=	100		101,886
deploy	474,654	474,654	=	-		-
pulse	29,052	29,052	=	36		-
ClaimRegistry						
acceptTransfer	85,778	85,778	=	228		-
delegate.exclusive.withdraw0	135,260	135,234	cd, v (−26)	420		-
delegate.exclusive.withdraw1	110,535	110,535	=	452		-
delegate.exclusive.withdraw3	265,028	265,028	=	516		-
delegate.nonexclusive	67,749	67,761	cd	420		-
delegate.nonexclusive.first_relay	84,849	84,861	cd	420		84,844
deploy	2,964,254	2,964,254	=	-		-
init	157,147	157,147	=	196		-
offerTransfer	77,410	77,410	=	260		-
registerCapture	136,887	136,913	cd, v (+26)	1,028		-
registerCapture.idempotent_repeat	45,712	45,712	=	1,028		-
registerFirst.after_registerCapture	135,489	135,489	=	1,476		-
registerFirst.tierE	226,692	226,682	cd, v (+26)	1,476		234,288
registerFirst.tierT	235,038	235,026	cd	1,476		-
registerMore.delegate	127,474	127,474	=	452		-
registerMore.owner	124,900	124,900	=	452		124,888
revokeDelegation	43,077	43,051	cd, v (−26)	260		-
withdraw	45,810	45,836	cd, v (+26)	228		-
DeviceRegistry						
confirm.executing_vote	73,595	73,595	=	36		-
confirm.first_vote	65,801	65,801	=	36		-
confirm.vote	48,805	48,805	=	36		-
deploy	1,332,460	1,332,460	=	-		-
enroll	104,167	104,167	=	196		-
finalizeRevocation.revoke	60,648	60,648	=	36		-
init	66,838	66,838	=	68		-
proveDoubleUse	98,482	98,494	cd	1,540		-
retireKey.first_relay	76,551	76,551	=	228		76,571
voteRevocation.last_vote	69,158	69,158	=	100		-
voteRevocation.opening	103,758	103,758	=	100		-
voteRevocation.vote	69,158	69,158	=	100		-

continued on the next page

Table R13 (continued)

Operation	EDR (gas)	Besu (gas)	Besu vs EDR	Calldata (B)	Floor	E10 mean (gas)
`Governance`						
`addValidator.executing_vote`	196,853	196,853	=	68		-
`addValidator.first_vote`	58,996	58,996	=	68		-
`addValidator.vote`	42,000	42,000	=	68		-
`deploy`	1,774,195	1,774,195	=	-		-
`removeValidator.executing_vote`	75,678	75,678	=	36		-
`removeValidator.first_vote`	58,544	58,544	=	36		-
`removeValidator.vote`	41,548	41,548	=	36		-
`setRelayer.executing_vote`	107,054	107,054	=	100		-
`setRelayer.first_vote`	59,128	59,128	=	100		-
`setRelayer.vote`	42,132	42,132	=	100		-
`setRole.executing_vote`	107,080	107,080	=	100		-
`setRole.first_vote`	59,306	59,306	=	100		-
`setRole.vote`	42,310	42,310	=	100		-
`HopRegistry`						
`anchorHopBatch`	96,460	96,460	=	68		80,118
`deploy`	886,099	886,099	=	-		-
`setHopCeiling.executing_vote`	380,904	380,904	=	292		-
`setHopCeiling.first_vote`	94,077	94,077	=	292		-
`setHopCeiling.second_version.executing_vote`	380,787	380,787	=	292		-
`setHopCeiling.second_version.first_vote`	94,077	94,077	=	292		-
`setHopCeiling.second_version.vote`	77,081	77,081	=	292		-
`setHopCeiling.vote`	77,081	77,081	=	292		-
`LabelPolicy`						
`deploy`	1,012,667	1,012,667	=	-		-
`labelOf.eth_estimateGas`	28,459	28,806	est.	-		-
`setRegime.executing_vote`	87,887	87,887	=	68		-
`setRegime.first_vote`	62,811	62,811	=	68		-
`setRegime.vote`	45,815	45,815	=	68		-
`setTable.executing_vote`	258,372	258,372	=	356		-
`setTable.first_vote`	72,704	72,704	=	356		-
`setTable.vote`	55,708	55,708	=	356		-
`StaticRules`						
`deploy`	1,315,714	1,315,714	=	-		-
`proveStaticViolation.rule1_lut`	118,407	118,419	cd	1,604		-
`proveStaticViolation.rule2_q50_without_prog`	109,337	109,337	=	996		-
`proveStaticViolation.rule2_q_gt_qpol`	109,349	109,349	=	996		-
`proveStaticViolation.rule3`	118,513	118,525	cd	1,604		-
`proveStaticViolation.rule4_D`	118,962	118,962	=	1,604		-
`proveStaticViolation.rule5_2_header`	123,597	123,597	=	1,828		-
`proveStaticViolation.rule5_6_rec_cutPre`	125,905	125,905	=	2,116		-
`proveStaticViolation.rule5_7_rec_cutPre`	125,396	125,384	cd	2,116		-
`proveStaticViolation.rule5_9_empty_prog`	105,186	105,186	=	772		-
`proveStaticViolation.rule6`	118,766	118,778	cd	1,604		-
`proveStaticViolation.rule7`	118,576	118,576	=	1,604		-
`proveStaticViolation.rule8_0_kernel_family`	128,185	128,185	=	1,604		-
`proveStaticViolation.rule8_1_sampling`	115,613	115,613	=	996		-
`proveStaticViolation.rule9_0_master_size`	114,712	114,724	cd	1,508		-
`proveStaticViolation.rule9_1_crop_area`	118,586	118,574	cd	1,572		-

[a] Measured in E17b (registered after E17, before its run) on a fresh $n = 4$ Besu devnet with `--tx-pool-max-tx-bytes` raised from the default 131,072 B to 1,048,576 B on every validator; the 289,028-byte transaction exceeds the default limit and was rejected in E17.

The twelve `BlockCourt.proveBlockFailure` rows of Table R13 carry the Besu gas measured in E17 and E17b; Table R14 gives the comparison with EDR. E17's classifier first priced calldata at the standard rate only; because each of three differences equals the EIP-7623 floor difference exactly, the floor rule was added before the results were finalised (disclosed). Court liveness on Besu at 65,535 declared MCUs therefore holds only when validators admit transactions of about 290 KB.

R10 Consensus, beacon and fork on Besu (E10)

Setup. Besu 26.8.1 QBFT with a 30-M gas limit at $n = 4$ and $n = 7$ (2-s blocks) and at $n = 7$ with 1-s and 5-s blocks; all validators on one 12-core host (2-GB container cap each); `netem` imposes an assumed inter-bloc round-trip matrix (5–320 ms, 0.5% loss per direction). The first-claim workload (W1 in Fig. R9) sends only first claims; the mixed workload

TABLE R11. E6 side by side. Pre-registered run and the disclosed post-fix rerun (same criteria, one code digest).

E6 number	pre-registered	post-fix
Honest in-scope renditions	10,577	10,577
Honest star failures (gate E6: must be 0)	**114**	**0**
Go class 5 unlinked / *U* flip up / *U* flip down	58 / 55 / 1	0 / 0 / 0
Star per profile, failures: P-AMZ, P-GMC, P-GMC/go	12, 11, 61	0, 0, 0
P-PIN, P-SHOP, P-SOC, P-TRELLIS	2, 25, 3, 0	0, 0, 0, 0
Go class 5 linked at 1000 × 750	28/86	86/86
Attack cases as expected (gate E6: all)	**35/38**	**38/38**
Joint aspect maximum (≤ 0.02)	0.015504	0.015504
Tier-X share without cooperation (served)	400/400	400/400 (reused)
Honest renditions above their source level	69	0
Renditions with ≤ 1 resample measured	8,603	8,661
luma above Λ_{chain} + 1, registered formula	46 (max/Λ_{chain} 1.555)	46 (1.555)
luma above Λ_{chain} + 1, footprint-certified only	–	0 (max/Λ_{chain} 0.506)
WF3 star P-AMZ bounded / off / on	888/895, 146/147, 147/147	895/895, 147/147, 147/147
WF3 star P-SHOP bounded / off / on	909/918, 151/153, 153/153	918/918, 153/153, 153/153
WF3 cascade P-AMZ bounded / off / on	361/596, 54/97, 50/97	363/596, 54/97, 50/97
WF3 cascade P-SHOP bounded / off / on	349/619, 55/103, 52/103	352/619, 56/103, 52/103
P-TRELLIS own ceiling: G3 / G2	254 / 45	252 / 47
Gate E6	**FAIL**	**pass (after three disclosed fixes)**
Re-derived from the same records with U printed only (the setting in force after gate E5):		
Honest star failures	**58** (P-GMC/go only)	0
Attack cases as expected	38/38	38/38
WF3 star P-AMZ / P-SHOP bounded	895/895, 918/918	895/895, 918/918
WF3 cascade P-AMZ bounded / off / on	299/596, 50/97, 50/97	299/596, 50/97, 50/97
WF3 cascade P-SHOP bounded / off / on	298/619, 50/103, 50/103	298/619, 50/103, 50/103
P-TRELLIS own ceiling: G3 / G2	299 / 0	299 / 0
Gate E6	**FAIL**	**pass (after three disclosed fixes)**

TABLE R12. E18: displayed pixels. Files displayed bit-exactly equal to Dec* on the certified region (of 400; 182 4:4:4 and 218 4:2:0) and the largest |Δ| (grey levels, RGB). Headless Chrome and the NumPy port skip one file above 4 MP.

Decoder (setting)	Exact on the certified region	Max |Δ| (grey)
AS9 family: libjpeg-turbo 1.2–2.1, IJG 6b (islow, fancy), Pillow, OpenCV (9 decoders)	400 each (3,600 pairs)	0
mozjpeg 1.0.1–4.0.3 (islow, fancy; 4 builds)	400 each	0
Headless Chrome; NumPy port of Dec*	399 of 399	0
IJG 7, 8, 8a–8d, 9 (islow, fancy; 7 builds)	182 (4:4:4 182/182; 4:2:0 0/218)	101
IJG 9a–9f (islow, fancy; 6 builds)	95 (4:4:4 95/182, max 1; 4:2:0 0/218)	101
turbo 2.1, 6b (islow, no fancy) / 9f (islow, no fancy)	182 / 95 (4:2:0 0/218)	101
ifast, any build, fancy or not	0	10 (turbo, 6b, fancy) to 101
float, any build, fancy or not	0	3 (turbo, 6b, fancy) to 101
Go image/jpeg	0	4 (4:4:4), 101 (4:2:0)

(W2) a listing mix (per 1,000 listings 250 first claims and 750 further claims, plus anchor and hop batches, delegations, a key retirement and court proofs). Offered rates 15–150 tx/s (15 tx/s only with 5-s blocks), three runs per point. The host was rebooted and its Docker engine upgraded (29.3.1 to 29.8.2) between the 2-s $n = 7$ cell and the period cells (Besu image unchanged).

Latency and knees (Fig. R9). Within capacity: median 1.90 s and 95th percentile 2.83 s at $n = 7$ (25 tx/s), 1.94/2.88 s at $n = 4$ (50 tx/s). The registered knee (highest rate with 95th-percentile latency within two periods and ≥ 99% included) is 50 tx/s at $n = 4$ for both workloads, 25 (first claims) and 50 (mixed) at $n = 7$; at $n = 7$, first claims, 50 tx/s, 98.5% were included. With 1-s blocks no rate met the criterion; with 5-s blocks the knee is 25 tx/s. Beyond the knee, latency grows to minutes and inclusion drops (61% at $n = 7$, first claims, 100 tx/s); the highest block-gas rate observed was 14.97 M gas/s. Three of six registered knee predictions failed as written: the mixed-workload knee was not higher than the first-claim knee at $n = 4$; the $n = 7$ first-claim knee was 25 instead of 50–100 tx/s; the 5-s knee was 25 instead of below 25 tx/s.

TABLE R14. E17: court rows on Besu. The 12 court rows on Besu 26.8.1 ($n = 4$) against EDR (receipt gasUsed). "calldata": the difference is the cost of one or two calldata bytes ("floor": at the EIP-7623 floor). The 65,535-MCU proof was rejected by Besu's transaction-pool size limit (default 131,072 B); E17b (registered after E17, before its run) raised only that limit to 1,048,576 B on every validator, and the proof was mined and emitted the expected failure.

Row	EDR (gas)	Besu (gas)	Δ (gas)	Class
`area_interior`	490,283	490,295	+12	calldata
`area_boundary`	408,832	408,832	0	equal
`triangle_6cells`	627,194	627,170	−24	calldata
`triangle9_probe`	741,639	741,639	0	equal
`boundary_mcu_with_fill`	452,620	452,620	0	equal
`deep_tree_wpC`	898,320	898,350	+30	calldata (floor)
`design_12mp_nD0`	994,330	994,330	0	equal
`design_12mp_nD256`	1,026,770	1,026,800	+30	calldata (floor)
`design_12mp_nD1024`	1,123,310	1,123,310	0	equal
`design_12mp_nD65535` (E17)	9,256,610	–	–	rejected: transaction size (289,028 B)
`design_12mp_nD65535` (E17b)	9,256,610	9,256,550	−60	calldata (floor)
`capture_24mp`	1,239,380	1,239,350	−30	calldata (floor)
`capture_200mp`	4,708,890	4,708,890	0	equal

Faults ($n = 7$, 50 tx/s). With $f = 2$ validators stopped for 240 s the chain kept finalising (68 blocks, longest gap 7 s); with $f + 1 = 3$ stopped for 180 s it halted (0 blocks) and produced its first block 260 s after the validators returned. A validator partitioned or crashed for 180 s caught up within 269 s and 261 s. No fault run produced a cross-node hash mismatch.

Beacon. On an idle chain without pulses, a predictor given the headers up to the previous epoch boundary ranked the true beacon first in 30 of 30 epochs; with pulses it never had the hash among its 4,096 candidates (0/30), nor with f sidecars absent (0/20), where the first honest pulse landed at least 14 blocks before the beacon height.

Fork. At $n = 4$ with $f + 1 = 2$ validator keys duplicated on both sides of a split, both sides finalised conflicting blocks from height 61, and the receipt verifier accepted a header of each side at height 144, each with three pinned seals. With one duplicated key the minority side committed no block. The control clause passes only after a disclosed post-run analysis fix: the frozen script counted blocks from a poll taken before the split containers answered; recomputed from the recorded heads (side B stays at the prefix head 60), the registered criterion is met; there was no rerun.

Receipt freshness. Within capacity a node's head trails the first node that saw a block by at most 685 ms at the 95th percentile (maximum 1.14 s); under overload the lag reached 110 s and during faults 442 s.

R11 Ledger ablation (E11)

Table R15 is the summary table of the ablation (main Sec. VIII-E); its footer gives D's measured inclusion latency on Besu (Table R17).

Model bound. On Besu, 14 of 100 censored repetitions exceeded the D-model's registered inclusion bound $(k + 1) \cdot 2$ s by at most 0.586 s (6 of 20 at $k = 0$ for each n, 1 of 20 at $k = 1$, none at $n = 7$, $k = 2$); all are within $(k + 2) \cdot 2$ s. Exit row 11 stays FAIL as written; the $(k + 2)$-period envelope is recorded as a disclosed revision of the model bound for the paper's model description, and the frozen bound, comparison and scores are unchanged.

R12 Costs, storage and timing (E12, E16, E16b)

Run 1 ran on a loaded shared host (all 60 lines flagged, median 1-min load maximum 6.42); the quiet rerun has 8 of 60 lines flagged (median 1.45), which were not repeated. Storage bytes are identical in both runs. The capture commit at 4 and 12 MP was slower in the quiet run; E16b (below) confirms the cause: OpenCV's thread pool under the one-core quota.

Profile (E16). One master per size, single thread, run while other jobs loaded the host (1-min load 3–7): ratios are reliable, absolute times indicative only. The "fast path" is the numpy Compile plus the libjpeg-turbo re-encode and compare; no native (C) verifier was built. The numpy Compile takes 74–88% of the fast path at 1024- and 2048-px listings, and the libjpeg-turbo encode 17.7–20.6 ms/MP there (22–31 at 512 px). Prediction (1) holds in part: Compile dominates, but the encode is about 20 ms/MP, not well below. A native verifier was not built; the encode alone indicates the order of magnitude of a native verifier's encoding step, not its total cost. Per 10^6 1024-px listings: fast path about 49–70 CPU-hours, the integer Python VerifyE 113–151, the encode alone 3.6–3.9. Prediction (2) fails: the 12-MP capture commit (6.0–6.6 s) is dominated by OpenCV's elliptical dilation of the certified region (3.8 s, of which `cv2.dilate` 3.58 s), with cell hashing 2.8 s, and a first-call gap appears only at 0.5 MP. OpenCV's thread pool is not pinned by `OMP_NUM_THREADS`.

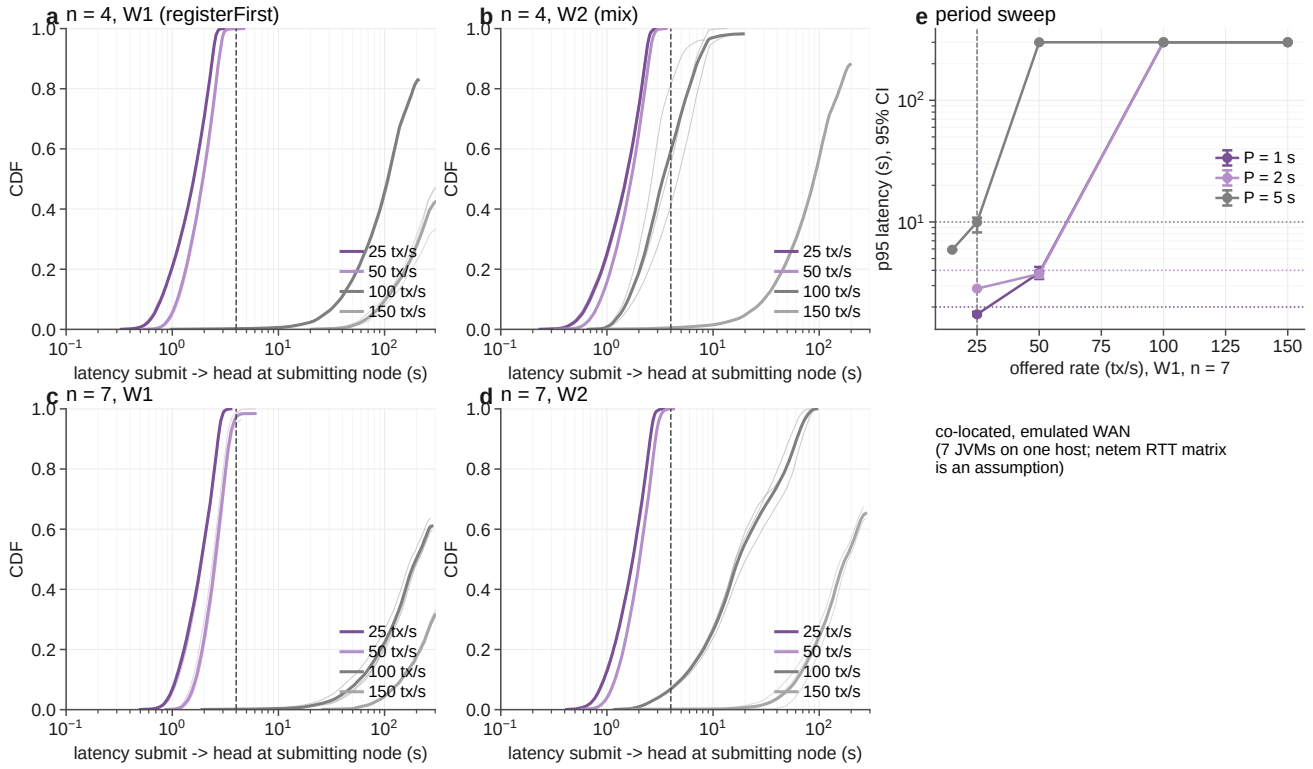


Fig. R9. E10 latency on the co-located Besu devnet with emulated WAN. (a)–(d) Latency from submission to the including head per offered rate, for $n = 4$ and $n = 7$ and the first-claim (W1) and mixed (W2) workloads; thin lines are single runs, the dashed line marks two block periods. (e) 95th-percentile latency (95% CI, capped at 300 s for display) against the offered rate for 1-, 2- and 5-s blocks ($n = 7$, W1).

TABLE R15. E11 ablation outcomes. Worst outcome over the adversary sweep ($k \leq f$ unless marked), $n = 4, 7$. B, C: full replayer [light client]. ✓ resisted, B bounded, R stated residual, ⊙ detected after gossip, × succeeds; † differs from the pre-registered prediction; ^{a} equals the amended prediction (grouped rows: any member). Sc5b: $k = f + 1$.

Scenario	A	B(10 s)	C	C'_{min}	D	D(f+1)	Collusion (D)	Detect. (s)	EDR
Sc1 split view	×	⊙ [×†]	✓ [×^{a}]	✓	✓	×	f+1	5.0 / 279	–
Sc2 conflicting grants	×	⊙ [×†]	✓ [×]	✓	✓	×	f+1	4.8 / 117	1/1
Sc2 theft, Sc2b-e, Sc2d', Sc2d"	×	✓ [×]	✓ [×]	✓	✓	✓	2f+1	–	10/10
Sc2f, Sc2g	×	✓ [×]	✓ [×]	✓	✓	✓	2f+1	–	6/6
Sc2f pre-proof aliases	R^{a}	R^{a}	R^{a}	R^{a}	R^{a}	R^{a}	0	–	1/1
Sc3(i) delay d	×	×	×	B	B	×	f+1	–	9/9
Sc3(iii) stockpiled beacons	×	×	×	✓	✓	×	f+1	–	1/1
Sc3(i)/(ii)/(iv)-(vii) thief-owned	×†	✓ [×†]	✓ [×†]	✓	✓	✓	2f+1	–	8/8
Sc3(vii) seller-owned	R	R	R	R	R	R	0	–	1/1
Sc4 equivocation	×	⊙ [×†]	×† [×]	✓	✓	×	f+1	4.3 / 157	1/1
Sc4' suppression	×	×	×	B	B	×	f+1	–	–
Sc5 forged proofs	×	✓ [×]	✓ [×]	✓	✓	✓	2f+1	–	6/6
Sc5b f+1 defamation (k = f+1 defamers)	×	×	×	×	×	×	f+1	–	8/8
Sc5t true failure, < f+1 checkers	✓^{a}	✓	✓	×	✓	✓	none	–	1/1
Sc6 G, E, L, D, H, P, T	×^{a}	×† [×^{a}]	×† [×^{a}]	✓	✓	✓	2f+1	–	10/10

Cost per first claim on D: 228,259 gas (registerFirst tier X + 1/64 anchorBatch). D latency on Besu QBFT (2-s period, n = 4, 7, k ≤ f censoring proposers, 100 repetitions; Table R17): median inclusion 1.64–1.92 s, max 5.08 s.

TABLE R16. Full E11 matrix. Observed worst outcome per scenario and back end, $n = 4/n = 7$; groups $k \le f$ and $k = f + 1$; B, C full replayer [light client]; D-EDR = EDR cells passing. Sc1–Sc5, Sc6 (ledger-rule cases G, E, L, D, H, P, T), Sc3-heldback and Sc2f-aliases are the pre-registered scenarios S1–S5, R4, R6D-heldback and R7E5E6-aliases of the artifact.

$k \le f$ Scenario	A	B(10 s)	B(600 s)	C	C'_{seq}	C'_{min}	C'_{native}	D	D-EDR
Sc1	×	⊙ [×†]	⊙ [×†]	✓ [×^{a}]	✓	✓	✓	✓	–
Sc2	×	⊙ [×†]	⊙ [×†]	✓ [×]	✓	✓	✓	✓	1/1
Sc2-theft	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2b	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2c	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	2/2
Sc2d	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	3/3
Sc2d’	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2d”	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2e	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/P1	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	2/2
Sc2f/P2	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/G1	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/P3	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2g	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc2f-aliases	R^a	R^a	R^a	R^a	R^a	R^a	R^a	R^a	1/1
Sc3(i)/seller	×	×	×	×	×	B	B	B	9/9
Sc3(iii)	×	×	×	×	×	✓	✓	✓	1/1
Sc3(i)/thief	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	–
Sc3(ii)	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc3(iv)	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc3(v)	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc3(vi)	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc3(vii)/thief	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	3/3
Sc3-heldback	✓†	✓ [✓†]	✓ [✓†]	✓ [✓†]	✓	✓	✓	✓	1/1
Sc3(vii)/seller	R	R	R	R	R	R	R	R	1/1
Sc4	×	⊙ [×†]	⊙ [×†]	×† [×]	×†	✓	✓	✓	1/1
Sc4’	×	×	×	×	×	B	B	B	–
Sc5	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	6/6
Sc5b/n4	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓^{a}	1/1
Sc5b/n7	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓^{a}	✓	✓	✓	7/7
Sc5t	✓^{a}	✓ [✓^{a}]	✓ [✓^{a}]	✓ [✓^{a}]	×	×	✓	✓	1/1
Sc6/G	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	3/3
Sc6/E	×	×† [×]	×† [×]	×† [×]	×†	✓	✓	✓	1/1
Sc6/L	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc6/D	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	2/2
Sc6/H	✓^{a}	✓ [✓^{a}]	✓ [✓^{a}]	✓ [✓^{a}]	✓	✓	✓	✓	1/1
Sc6/P	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1
Sc6/T	×	✓ [×]	✓ [×]	✓ [×]	✓	✓	✓	✓	1/1

$k = f + 1$ Scenario	A	B(10 s)	B(600 s)	C	C'_{seq}	C'_{min}	C'_{native}	D	D-EDR
Sc1	–	–	–	⊙ [×^{a}]	×	×	×	×	–
Sc2	–	–	–	⊙ [×]	×	×	×	×	1/1
Sc2-theft	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2b	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2c	–	–	–	✓ [×]	✓	✓	✓	✓	2/2
Sc2d	–	–	–	✓ [×]	✓	✓	✓	✓	3/3
Sc2d’	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2d”	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2e	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/P1	–	–	–	✓ [×]	✓	✓	✓	✓	2/2
Sc2f/P2	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/G1	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2f/P3	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2g	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc2f-aliases	–	–	–	R^a	R^a	R^a	R^a	R^a	1/1
Sc3(i)/seller	–	–	–	×	×	×	×	×	9/9
Sc3(iii)	–	–	–	×	×	×	×	×	1/1
Sc3(i)/thief	–	–	–	✓ [×]	✓	✓	✓	✓	–
Sc3(ii)	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc3(iv)	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc3(v)	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc3(vi)	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc3(vii)/thief	–	–	–	✓ [×]	✓	✓	✓	✓	3/3
Sc3-heldback	–	–	–	✓ [✓†]	✓	✓	✓	✓	1/1
Sc3(vii)/seller	–	–	–	R	R	R	R	R	1/1
Sc4	–	–	–	⊙ [×]	×	×	×	×	1/1
Sc4’	–	–	–	×	×	×	×	×	–
Sc5	–	–	–	✓ [×]	✓	✓	✓	✓	6/6
Sc5b/n4	×	×	×	×	×	×	×	×	1/1
Sc5b/n7	×	×	×	×	×	×	×	×	7/7
Sc5t	–	–	–	✓ [✓^{a}]	×	×	✓	✓	1/1
Sc6/G	–	–	–	✓ [×]	✓	✓	✓	✓	3/3
Sc6/E	–	–	–	×† [×]	×†	✓	✓	✓	1/1
Sc6/L	–	–	–	✓ [×]	✓	✓	✓	✓	1/1
Sc6/D	–	–	–	✓ [×]	✓	✓	✓	✓	2/2
Sc6/H	–	–	–	✓ [✓^{a}]	✓	✓	✓	✓	1/1

TABLE R17. E11 back end D on Besu QBFT. Inclusion delay with k censoring proposers (20 repetitions) vs. the D-model bound $(k+1)\cdot 2$ s.

n	k	Reps	Median (s)	p95 (s)	Max (s)	Bound (s)	>Bound
4	0	20	1.64	2.35	2.35	2	6
4	1	20	1.87	4.22	4.22	4	1
7	0	20	1.71	2.59	2.59	2	6
7	1	20	1.86	4.01	4.01	4	1
7	2	20	1.92	5.08	5.08	6	0

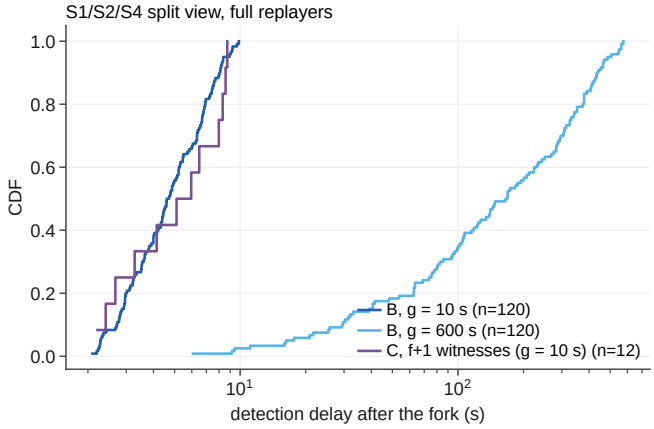


Fig. R10. E11: equivocation detection. Back end B's detection time against the gossip period.

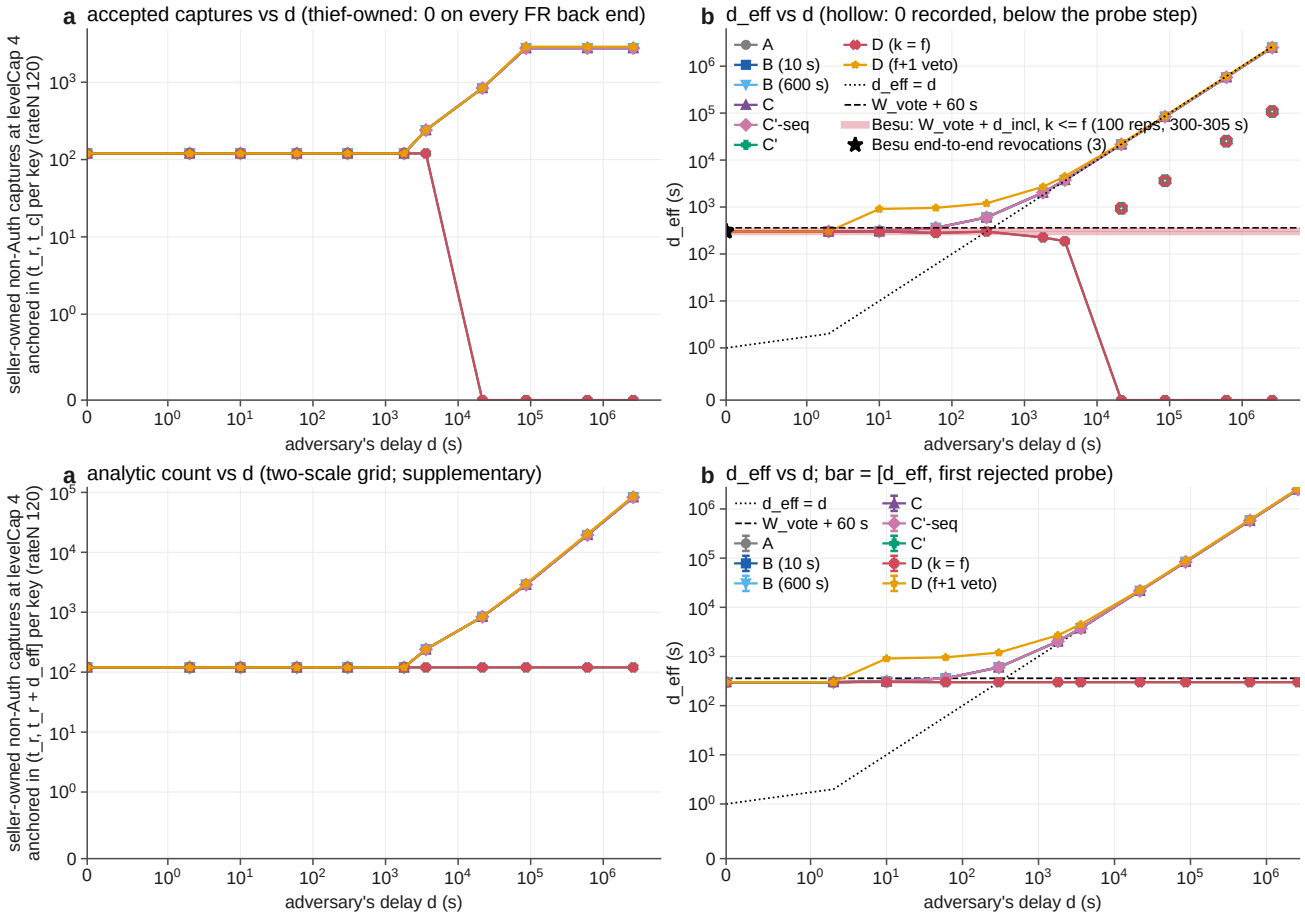


Fig. R11. E11 Sc3 backdating window against the operator delay. Top: pre-registered probe grid (with its resolution limit). Bottom: the two-scale grid applied after the freeze (disclosed), which gives C′ and D a window of 303–310 s at every delay.

TABLE R18. E12 costs and storage. Quiet re-run (disclosed deviation: run 1 ran on a loaded shared host, every line load-flagged; both are reported). Medians over masters of per-master medians of 1-thread runs; times in ms unless marked. Fast = numpy Compile plus libjpeg-turbo re-encode (no native verifier was built); its design target ≤ 20 ms/MP is a reported comparison. Capture at 4-12 MP is inflated by OpenCV threading under the one-core quota (E16b: 2.95 s at 12 MP with one thread). "Upscaled masters are Lanczos-upscaled natives. Evaluate excludes PubSeg inference (priced by E5). Load-average flags (> 2.0): 8.

Master	Listing (px)	*n*	Capture (s)	Compile (ms)	X0 enc. (ms)	VerifyE py (ms)	Fast (ms)	Fast (ms/MP)	Evaluate (ms)	Hop (ms)	Opened (KiB)	Bundle (KiB)	*U* rec (KiB)	Store/cap (MiB)	Store/list (KiB)
0.5 MP	512	8	0.08	48.2	4.5	140.3	63.1	289.5	157.8	60.0	374.0	414.4	49.0	1.41	497.2
0.5 MP	1024	2	0.08	95.3	9.6	282.8	124.8	274.3	320.7	105.1	562.8	654.3	111.5	1.41	855.0
1 MP	512	9	0.17	31.3	4.1	100.5	39.6	294.3	111.8	35.4	246.3	285.2	32.0	2.82	360.2
1 MP	1024	5	0.17	126.7	10.1	358.8	162.5	307.1	398.2	108.4	792.8	903.1	119.7	2.82	1155.6
2 MP	512	9	0.60	45.5	4.4	132.9	57.5	277.2	147.9	56.9	268.0	299.7	46.8	4.34	373.8
2 MP	1024	9	0.60	181.9	13.3	480.4	219.5	289.9	539.3	153.6	887.0	980.3	170.7	4.34	1240.4
4 MP	512	10	1.61	41.0	4.2	129.5	52.0	283.9	143.4	60.0	269.4	302.4	47.9	7.62	390.4
4 MP	1024	9	1.61	148.3	13.1	458.8	187.9	307.2	512.2	147.8	1288.7	1367.1	181.6	7.62	1637.1
4 MP	2048	4	1.61	573.4	43.7	1578.2	699.4	235.7	1805.7	308.4	1532.1	1785.5	698.8	7.62	2585.5
8 MP"	512	9	4.83	43.7	5.3	129.6	56.1	268.3	145.1	55.2	229.8	268.5	49.0	23.16	353.8
8 MP"	1024	9	4.83	147.4	13.3	430.3	189.0	260.1	484.5	146.7	789.2	889.0	177.5	23.16	1163.8
8 MP"	2048	10	4.83	585.2	51.5	1679.2	742.4	257.0	1936.7	310.9	2941.5	3253.8	702.9	23.16	4335.6
12 MP"	512	7	9.80	34.4	4.0	109.8	46.5	292.1	125.3	55.7	273.1	307.8	49.0	25.54	378.5
12 MP"	1024	8	9.80	148.2	12.5	440.6	189.1	291.0	495.7	147.6	1060.4	1148.4	176.8	25.54	1366.2
12 MP"	2048	9	9.80	617.4	41.8	1861.7	768.9	286.8	2109.7	294.9	3908.0	4149.5	658.2	25.54	4860.0

TABLE R19. E12 quiet run and loaded run. Quiet run (primary) and run 1 (loaded host, disclosed) for selected cells: medians in ms unless marked.

Master	Listing (px)	Run	Capture (s)	Compile	VerifyE py	Fast	Fast (ms/MP)	Evaluate	Hop
0.5 MP	512	quiet	0.08	48.2	140.3	63.1	289.5	157.8	60.0
0.5 MP	512	run 1	0.09	50.7	144.0	63.2	297.5	162.6	52.8
1 MP	1024	quiet	0.17	126.7	358.8	162.5	307.1	398.2	108.4
1 MP	1024	run 1	0.18	137.3	385.7	176.7	327.7	438.5	117.8
4 MP	1024	quiet	1.61	148.3	458.8	187.9	307.2	512.2	147.8
4 MP	1024	run 1	1.12	164.0	520.0	205.7	324.4	584.2	164.4
4 MP	2048	quiet	1.61	573.4	1578.2	699.4	235.7	1805.7	308.4
4 MP	2048	run 1	1.12	661.6	1735.2	772.2	260.7	1964.2	366.3
12 MP	1024	quiet	9.80	148.2	440.6	189.1	291.0	495.7	147.6
12 MP	1024	run 1	5.40	211.6	618.8	268.1	327.8	696.0	190.4
12 MP	2048	quiet	9.80	617.4	1861.7	768.9	286.8	2109.7	294.9
12 MP	2048	run 1	5.40	734.4	2129.3	916.6	354.4	2484.7	326.6

TABLE R20. E16 fast-path profile (one master per size, loaded host).

Master	Listing (px)	Canvas (MP)	Fast path (ms)	Compile (ms)	libjpeg encode (ms)	Encode (ms/MP)
4 MP	1024	0.74	250.5	207.6	14.0	18.8
4 MP	2048	2.90	920.7	810.6	51.3	17.7
12 MP	1024	0.63	176.3	129.7	12.9	20.6
12 MP	2048	2.43	1113.8	954.1	44.6	18.4
12 MP	512	0.17	54.6	39.0	4.3	26.2

E16b: quiet host and thread test. Registered after E16's result and committed before its run, E16b re-ran E16 alone on a quiet host (5-min load below 1.0 at the start; no line above a 1-min load of 2.0) under the same one-core CPU quota as E12 and E16, and timed the 12-MP capture commit and the dilation alone six times each with OpenCV's default 12 threads and with one thread (ABBA order). Quiet fast path at listings ≥ 1024 px: 223–366 ms/MP (E16 under load: 282–458); libjpeg encode 17.1–19.3 ms/MP, 5–9% of the fast path. Capture commit at 12 MP: 9.78 s with default threads (E12 quiet: 9.80 s); default vs one thread 9.99 vs 2.95 s (3.4×), dilation 7.66 vs 0.80 s (9.6×), with a median 84.8 s of throttled CPU time per capture against ≈ 0. The registered criterion (ratios ≥ 1.25 with more throttling) confirms the threading explanation: under a one-core quota the 12-thread pool is throttled and burns CPU in parallel overhead, and on a loaded host fewer of its threads run at once. Without a quota, 12 threads take 2.38 s and one thread 2.94 s. The single-thread 2.95 s is the comparable one-core cost of a 12-MP capture commit; the 4–12-MP capture commits of E12 and E16 are artefacts of this interaction.

R13 Worked example of a certified listing

Fig. R12 shows, on a schematic G3 listing, which MCUs are certified, which are declared and where the drawn silhouette is eroded.

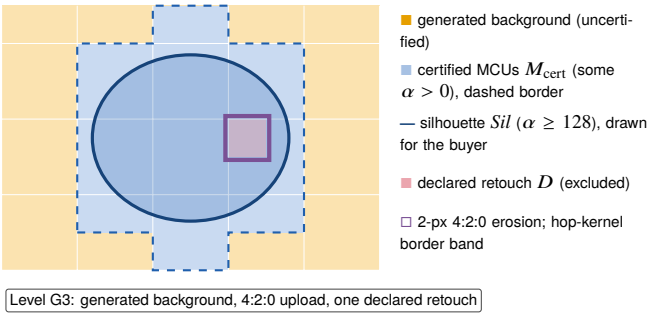


Fig. R12. Worked example of a G3 listing (schematic; one square is one 16×16 MCU; main Fig. 2 shows a real example). The generated background and D are not certified. The drawn silhouette excludes D and, at 4:2:0, is eroded by 2 px along borders between certified and uncertified MCUs (violet frame; here only the border with D, since Sil stays inside the outer certified border); after a resampling hop, pixels whose kernel footprint leaves M_{cert} are not bounded (main Proposition 1).

R14 System architecture

Fig. R13 shows CUTLINE end to end; the numbered arrows follow the workflow of main Sec. IV (1–2 capture, 3–5 compile and upload, 6–7 hops, 8 adjudication, 9 evaluation).

R15 Ledger details

This section expands main Sec. VII. **Consortium parameters.** Besu QBFT with $n = 7$ validators in seven regulatory blocs, at most one seat per bloc (the EU counts as one), 2-s blocks and a block gas limit of 30 M gas; at $n = 4$ any two colluding blocs can break safety or censor. We use an EVM chain for off-the-shelf tooling, the P-256 verification precompile and a programmable court. The nine registry contracts are each below 24,576 bytes.

Court, path (a): single-MCU proof. For a 4:4:4 tier-X claim the proof carries one failing MCU with its coefficients and R_Y path, the covering cells (at most 3×3 by static rule 3) with their paths, the background leaf if the block's minimum alpha is below 255, and the program. The court lets receipt-only parties (customs, small platforms) see FAILED and makes one honest watcher's verdict binding everywhere, in the any-trust sense of Arbitrum [5]. Equivalence of the contract with the reference predicate (AS10) was tested on 22,306 cases (main Sec. VIII-E). A proof costs about $21{,}000 + 40$ gas per non-zero calldata byte under the EIP-7623 floor [6], at most 1.0 M gas for captures up to 12 MP with at most 45 declared retouched MCUs, and 14,349,680 gas at the adversarial size bound of static rule 9, below the 16,777,216-gas per-transaction cap [7], so gas never blocks a proof. Transaction size can: that bound case (measured on EDR only) carries 416,324 calldata bytes, so validators must admit transactions above Besu's default pool limit of 131,072 bytes. With the limit raised to 1,048,576 bytes, the 289,028-byte proof of the 12-MP case with 65,535 declared MCUs (E17b) was mined on Besu (§R9). Violations of static rules 1–9 are proven in $O(1)$: the proof names the rule and the offending entry and costs 105,186–128,185 gas including the strike.

Court, path (b): tallied verdicts. Validators run the pinned checker on claims the court cannot check (4:2:0, fallback tier) and vote transcripts within $W_{\text{vote}} = 150$ blocks under the tally rule of main Sec. VII; a defamation by $f + 1$ colluders is cleared by appeal at $n \geq 7$. Uploads at 4:2:0 are capped at G3.

Alternative: signed captures and transcode manifests. Camera signing [8] plus platform-signed C2PA transcode actions (`c2pa.transcoded` [9]) need no openings or ledger but trust every transcoder, leave region and composite self-declared, and keep status per validator [10].

Status and receipts. The ledger cap is 0 for an unknown claim or a key invalid at the anchor, 1 if the claim's principal is struck, the claim is withdrawn or its claimant is no longer authorised, 2 if the capture is time-unverified, and 4 otherwise. Parties without a node use receipts (a header with $2f + 1$ commit seals plus storage proofs), accepted only within 30 blocks of a head reported by $f + 1$ validators and within 5 min of wall-clock time; a stale receipt yields "status unknown" (G2). A receipt-only party that does not run VerifyE never shows a G3/G4 label.

Labels. Labels are per-jurisdiction tables, monotone in the level, that pin G0/G1 to the regime's "no provenance information" code.

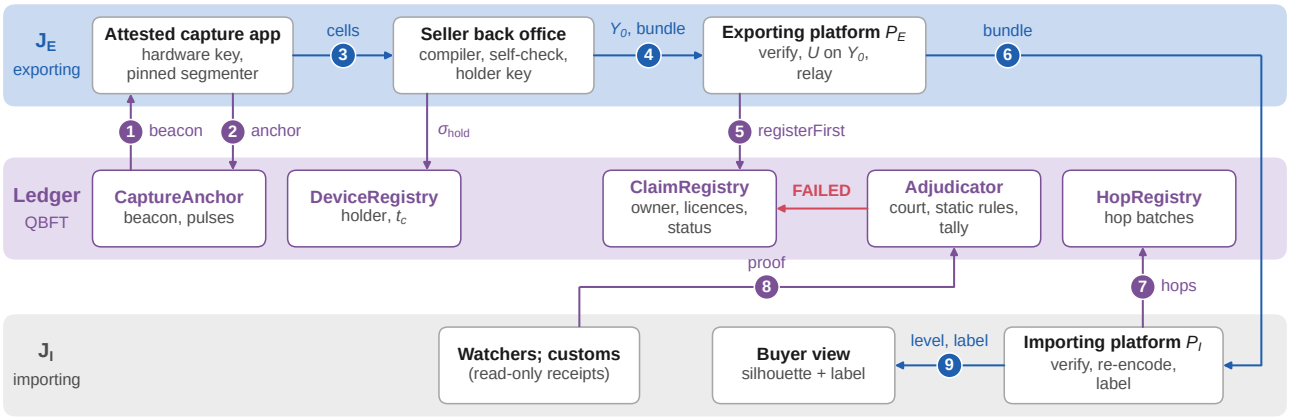


Fig. R13. CUTLINE architecture. Top: the exporting jurisdiction J_E; middle: the consortium ledger (one validator per regulatory bloc); bottom: importing jurisdictions J_I. Only roots and status go on chain; the claim bundle (product cells at most 2× listing resolution and the public background) goes to P_I; masters, unopened cells, salts and identities never leave J_E.

Privacy. On chain there are only salted or keyed digests, per-capture owner pseudonyms and status; since salted hashes count as personal data [11], erasure deletes the off-chain salts and bundles. Masters, unopened cells and identities never leave J_E; all captures of one device remain linkable.

Costs. The canonical first-claim cost (capture registration plus claim, evaluated build) is 226,682 gas on Besu, about 132 per block of 30 M gas by gas arithmetic. Variants of the same operation: 226,668–226,692 on EDR (calldata bytes and the signature's v value differ); 234,288 on average under the E10 load (Table R13); and 228,259 in the ablation summary, which adds a $1/64$ share of an anchor batch (Table R15).

R16 Positioning: full table

Table R21 places CUTLINE against the closest lines of work.

TABLE R21. Positioning and breaks by design. Published costs are quoted with their page or table.

Line of work	Reference needed at verification	Lossy codec	Partial disclosure	Adversarial guarantee	Re-encode tolerance	Cost (as published)	E4 (this paper)
Semi-fragile / content-based signatures: Lin–Chang [12] (abs); Sun–He–Tian dc features [13]	no (robust features)	yes, by feature invariance	no	feature-level: "more robust … less secure" [13, abstract]; PhotoProof's comparison lists non-negligible error for this class [14, Table I p.3]	built into the features	cheap	–
PhotoProof [14]	no (SNARK over a signed original)	no	no (whole image)	cryptographic, for a configurable set of edits	none	128 × 128: prove 306 s, verify 0.5 s, proving key 2.6 GB [14, Table II p.12]	breaks by design (no lossy codec)
ZK-IMG [15]; VIMz [16]; HyperVerITAS [17]; VerITAS [18]	no (proof)	no; HyperVerITAS/VerITAS bound rounding error	hiding (ZK)	cryptographic	rounding only	ZK-IMG HD crop 7.8 s; up to 617 s and 102 GB end to end when only the input is hashed, and up to 2,236 s and 308 GB when input and output are hashed (the committed-content case comparable to CUTLINE) [15, p.10]; VIMz 33 MP resize 49 min, 9.5 GB [16, Table 8 p.13]; HyperVerITAS 33 MP 6.6 min, 27.4 GB [17, Table 4 p.27]	breaks by design (no lossy codec)
Redactable / croppable signatures [19] (abstract); [20] (abstract) as described in [21, p.2]; [21]	the retained blocks	no: verification needs decompression [21, p.3]; a "bandwidth-efficient homomorphic signature scheme supporting both cropping and compression remains an open problem" [21, p.2]	yes (subset)	cryptographic	none	O(1) aggregated BLS signature and traffic [21, pp.1-2]	breaks by design (verification needs decompression)
Compression-tolerant signature: Erfurth [22]	no (a third party derives a valid signature after recompression, without the key)	yes, but only for quantisation tables of powers of two [22, p.1]	no (no cropping [21, p.2])	cryptographic (unforgeability proved, pp.6-7)	recompression with power-of-two tables	32.5 kb signature [22, p.9]	–
Proactive localisation watermarks [23, 24]; C2PA soft binding [25]	no (embedded)	yes (robust watermark)	n/a	learned, not adversarial: TrustMark bit accuracy about 0.95, removable [25, p.8, pp.1, 4]	yes	EditGuard 37.8 dB at 512 × 512 [23, Table 2 p.6]	EditGuard: not run (no licence granted); OmniGuard: not run (no licence granted); TrustMark: measured in E4
C2PA hard binding [9]	the manifest	breaks on re-encode	no	signature on self-declared actions [9, pp.188-191]	needs transcode manifests	cheap	breaks by design (no demo: c2pa not installed)
CUTLINE	opened product cells (+ public fill); not homomorphic	yes: exact coefficient test (`X0`) for IJG-style tables of any values within Q_{pol}; η-cell (`X1`)	yes: capture-fixed, occupancy-complete region	cryptographic for the region's pixels; empirical for product coverage (`AS7`)	verified hops, zero excess for `X0` chains	not bandwidth-efficient: 793 KiB of openings for a 1024-px listing from a 1-MP master (E12; about 3.0 MB per certified MP in a design probe); `X0` verifier 359 ms for that listing in Python (E12); court ≤ 1.0 M gas (v1, captures ≤ 12 MP with ≤ 45 declared MCUs)	–
gnark stand-in tile (our feasibility probe)	no (proof)	yes (JPEG relation)	yes	cryptographic	n/a	0.83 s per 32 × 32 tile, about 14 min per 1024 × 1024 listing (our measurement)	cost row only

R17 Evaluation narrative (previous full text of the manuscript's evaluation)

The manuscript's evaluation (main Sec. VIII) was condensed around its Figs. 3–5. This section keeps the previous full text of that section, unchanged in substance, so that every protocol detail, number, failure and disclosure that the main text now only summarises remains findable; references to "Sec." and "Table" without "main" point into this supplement or, via "main", into the manuscript. Table R22 is the condensed E4 table that the manuscript printed before the restructuring; Fig. 3b of the manuscript plots the same numbers.

R17.1 Setup, pre-registration and artifact

Data. 600 ABO catalogue images [26], 200 Objectron phone-video frames [27] and 200 CO3D frames [28] with SHA-256 manifests. Masters are decoded frames and catalogue images, not phone stills, and the app is emulated; coverage (E5) may differ on phone stills. **Platforms.** Emulated ladders P-AMZ (Amazon's served table), P-SHOP, P-GMC, P-PIN, P-SOC and P-TRELLIS. **Statistics.** One-sided 95% Clopper–Pearson (CP) bounds; zero-failure completeness uses 299 listings per stratum (upper bound 0.997%); paired comparisons use the exact conditional McNemar test [29] with Holm's correction [30]; clustered rates use Kish's design effect [31]. **Pre-registration and artifact.** Every experiment block was pre-registered with its criteria and frozen in a commit of the project's private repository before its first full run (internal commit timestamps; §R1.3). Additional experiments E13–E21, registered after the main results and each before its own run, are marked as such. The code is released with this report at `https://github.com/yebof/cutline`. Table II lists every registered outcome.

R17.2 Completeness, soundness and adaptive attackers (E1–E3)

Completeness (H1). 17,940 honest listings in 60 strata (299 each) produced 0 certified-block failures and 0 container rejections: false-rejection rate (FRR) $\leq 0.997\%$ per stratum, $\leq 0.0167\%$ pooled. Per class: X0 0/11,661, float-DCT (class 3) 0/3,588, Go (class 5) 0/897, pinned XP 0/1,794 (Supplement Table R3). mozjpeg 4.0.3's default tables violate static rule 2, so such uploads end at G1.

Soundness (H2). Six attack classes (A1 defect removal, A2 text, A3 logo swap, A4 recolour, A5 retouch, A6 expansion) in six upload profiles, $n = 198$–200 per cell: detection 1.0 in all 36 cells (lowest lower bound 0.985), 0 honest rejections, 0 violations of Theorem 2(e) over 7,136 forced listings, and policy attacks never reached G4 (20/20).

Adaptive attackers (H3). An attacker that places each edit optimally inside the tolerant cell at $\eta = 0.25$ moves to the screen a worst-cell median of 3.3% and a worst-cell 90th percentile of 8.1% of the edit; under X0 it moves nothing (Fig. R3).

R17.3 Baselines (E4, H4)

Protocol. Each method is calibrated on 299 honest listings per upload profile (U85: a seller q85 upload; AMZ: after the P-AMZ re-encode) at a false-positive rate (FPR) of 1% and tested on 299 further honest listings and the attacked ones (Table R22): tier T, Sun–He–Tian semi-fragile signatures [13], a signed thumbnail, pHash, PDQ [32], a Birthmark-style replay [33], a PROVES-style check [34], TruFor [35], Mesorch [36] and TrustMark [25], plus 150 Stable-Diffusion attacks [37]: a logo swap and a texture change inside the silhouette (A3sd, A7sd) and an expansion ring just outside it (A6sd). EditGuard, OmniGuard (no licence) and CAT-Net (no weights) could not be run. H4 is falsified iff a baseline dominates CUTLINE (higher TPR, exact McNemar, Holm $p < 0.05$) on any class, or CUTLINE convicts an honest listing.

Outcome. H4 is not falsified: the exact test detected 100% of every class on both profiles and convicted 0 of 299 honest test listings on each (CP upper bound 1.0%); over 189 McNemar comparisons there are 0 dominations and no higher baseline TPR (equal in 5). The comparison is contextual: a verifier holding the reference must detect every edit that survives the codec (Remark 1); TruFor and Mesorch see no reference, and TrustMark localises nothing by design. Sun–He–Tian and the thumbnail hold the same reference (†) but are coarse. Two tolerant tests on the same openings were registered after E4, each before its run (E19, E21; §R4). A global-threshold pixel test (‡) flags a listing if any certified pixel is off by more than T (91 at U85, 148 at AMZ; FPR 1%): it detects 0–96% per class at U85 and 0–62% at AMZ, below CUTLINE (exact McNemar, Holm $p < 0.05$) on 17 of 18 cells, with no significant difference on A6sd at U85 (96%, $p = 0.5$). The X1 cell test (§), which compares every certified coefficient with its own cell (η calibrated to 0.178 at U85 and 0.181 at AMZ, FPR 1%; at AMZ against the hop reference rebuilt from the openings), detects 100% of all 18 cells ($p = 1$ against CUTLINE) with 1/299 and 0/299 honest alarms: every attacked listing needs $\eta \geq 1.0$, every honest one at most 0.183. With equal reference information, a per-cell tolerant test thus detects what the exact test detects on E4; the advantage over the baselines comes from the opened capture and the transcoding-aware reference, not from exactness. Exactness adds a one-point acceptance set with no slack parameter (Remark 1) and zero adversarial excess, where the $\eta = 0.25$ cell lets an optimal attacker move a worst-cell median of 3.3% of an edit to the screen (H3). The A6sd ring (radius 21–64 px, outside *Sil*) changed at least 1,560 certified pixels of the dilated region A in all 50 listings and was detected 50/50. Generation wholly beyond the certified MCUs (RF2) is not detected: in E13 (§R4) all 30 such diffusion listings, a median 13.4 px from the product, kept G3 (Fig. R5). **Tier T.** With

TABLE R22. Condensed E4 table (previous manuscript version). Detection (TPR %) at a matched false-positive rate of 1% (U85 upload profile; after the Amazon re-encode CUTLINE tier X is again 100.0 on every class). The comparison with reference-free methods is contextual.

Method	A1	A2	A3	A4	A5	A6	A3sd	A6sd	A7sd	FRR
CUTLINE tier X (X0)	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	0.0
tier T (4:4:4 table)	86.9	91.9	100.0	68.7	91.9	89.9	–	–	–	1.0
X1 cell, calibrated[§]	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	0.3
Pixel, global max[‡]	27.9	42.4	89.9	11.2	0.0	4.0	30.0	96.0	54.0	0.0
Sun–He–Tian Δ=4[†]	67.4	79.8	100.0	61.2	80.8	73.7	92.0	0.0	96.0	0.0
Thumbnail[†]	61.6	73.7	100.0	60.2	78.8	67.7	94.0	100.0	96.0	0.0
pHash	0.0	1.0	2.0	15.3	0.0	0.0	0.0	38.0	2.0	0.3
PDQ	0.0	2.0	13.1	24.5	0.0	0.0	2.0	78.0	10.0	0.0
Birthmark replay	7.0	3.0	23.2	32.7	2.0	14.1	14.0	90.0	40.0	0.0
PROVES-style	0.0	0.0	1.0	6.1	0.0	0.0	0.0	0.0	0.0	0.0
TruFor, Mesorch	0.0 on every class									0.3, 0.0
TrustMark[b]	0 on every class (no localisation)									0.0

[b]By design: TrustMark is a soft binding and does not localise edits. [†]Hold the same reference as CUTLINE; Δ=8, 16: 42.4–100.0 on every class but A6sd (0.0). [§]E21: every certified coefficient against its cell, η=0.178 (AMZ: 0.181, 100.0 on every class, 0/299 honest). [‡]E19: maximum certified-pixel difference from the reference, T=91; at AMZ (T=148) 0.0–62.0. A6sd: tier-test verdict on the ring. All Δ, the AMZ rows and the E19 variants: Tables R4 and R5.

tables calibrated by an order-statistic rule [38], tier T lets threshold-hugging edits (A10) through at a median 10.7–79.6% of the edit, against 0.1–3.1% for the X1 cell at $\eta = 0.25$ and 0 under X0, hence its G2 cap.

R17.4 Region coverage and staging (E5, H5)

Protocol. 300 masters (100 ABO white-background products, 100 ABO products composited into scene backdrops and 100 DIS5K images [39]) were captured with BiRefNet-lite [40] and listed in four honest workflow types: studio, outdoor and prop-rich lifestyle backgrounds pasted from 140 frozen generated scenes, and a product-conditioned inpainting arm (pc, 200 listings). Gate E5 requires (a1) per honest type a false-alarm upper bound of $U > \tau_U$ of at most 15% (Kish-adjusted for the three pasted types, plain CP for pc), among *complete* listings whose silhouette holds at least 98% of the ground-truth product, and (a2) per mask-only staging type a detection lower bound $\geq$ 90%; otherwise U leaves the level criteria (written before the run). The 15% target, $\tau_U = 32$ and the form of U were fixed on a 29-listing pilot, before the run.

Outcome: gate E5 fails, so U is printed only. (a1) fails on all four types, with false-alarm rates 0.287 (studio, 77/268), 0.310 (outdoor), 0.590 (lifestyle) and 0.820 (pc, 146/178); the denominators exclude the listings of the 32 masters (22 pc listings) whose capture missed at least 2% of the product by ground truth, which a deployed app does not have. (a2) holds on every staging type (detection 0.952–0.985, lower bounds $\geq$ 0.943; Supplement Fig. R6). The false alarms are mostly touching props or shadows and slivers. **Coverage against ground truth (E14).** A mean 4.7%, 4.5% and 0.39% of the product lies outside *Sil* (white, scene, DIS; median 0), and 20, 8 and 4 of 100 masters miss more than 2% (upper bounds 0.277, 0.140, 0.089). These misses are an operational residual of every label: product parts outside *Sil* are uncertified, and content generated there is not detected.

A projected-gradient attack [41] on the generated fill evades U on 44/100 staged listings at $\epsilon = 2/255$ and 55/100 at $4/255$ without changing a certified pixel (*surrogate white-box*: the canonical gate failed; transferred to BiRefNet-lite, 2/100).

Honest workflows. A workflow family is deployable iff at least half of it reaches its target level. WF1 (constant backgrounds, G4) passes with 300/300 listings and WF2 (lifestyle and product-conditioned backgrounds, G3) with 500/500, but only because U is printed only (150/500 with U a level criterion). For WF3 (re-hosting), renditions through P-AMZ and P-SHOP keep the source level on one-hop ladders (440/440, 447/447) but on cascades only 150/290 (0.517), 150/297 (0.505) and 298/619 (0.481), because renditions resampled twice keep the registered chain cap G2: WF3 is deployable on one-hop ladders of aspect-preserving resizes only (Table R9); a hop that crops to another aspect ends at G2 (Sec. VIII-D).

R17.5 Verified hops and platform ladders (E6, H6)

The pre-registered run failed gate E6 and falsified H6 as written: 114 of 10,577 honest star-ladder renditions missed their level (58 Go class-5 renditions unlinked, 56 U flips), 3 of 38 attack cases were off by one level, and 46 of 8,603 renditions had a luma deviation from the direct reference above $\Lambda_{\text{chain}} + 1$ (the printed bound is luma, Proposition 1(iii)).

Three root causes, none a theorem violation (Go's chroma edge padding, U re-tested on rendition grids, border pixels whose kernel footprint reaches uncertified background), were fixed; a single rerun with the same criteria gives 0 honest failures, 38/38 attack cases and 0 footprint-certified pixels with a luma deviation above $\Lambda_{\text{chain}} + 1$ (maximum $0.51\Lambda_{\text{chain}}$, Fig. R7): gate E6 passes after three disclosed post-registration fixes, on the diagnostic data. A pre-registered confirmatory

rerun (E15) on 50 fresh ABO masters passes: 0 of 3,603 honest renditions fail, 38/38 attack cases, WF3 star 185/185 (cascades 0.37–0.39). Every rogue, squashed, nine-hop or withheld-ancestor chain was unlinked or capped as predicted. A fixed 4:3 crop (our test template, not a modelled platform size) cuts certified MCUs from every product not near 4:3, so 255 of 269 renditions at P-AMZ 500 × 375 end *partial* (G2), square ones too (26/26; E20); non-MCU-aligned output sizes do not lower the level (0 of 3,607 partial). A trellis platform is "coarse" (G2) under the default hop ceiling.

R17.6 Court, gas, consensus and ledger ablation (E9–E11)

E9. The court equals the reference package on all 22,306 cases on Besu 26.8.1 as on EDR. Of 121 gas rows, 108 match EDR up to calldata bytes or the signature's v, and 11 of the 12 court rows copied from EDR reproduce (E17); the 289,028-byte proof with 65,535 declared MCUs exceeds Besu's default pool limit and was mined only once it was raised (E17b; Table R14). All 53 regression attacks revert. On 202 accepted container files, eight parsers and decoders (three libjpeg builds, Pillow, OpenCV, Go, headless Chrome, our parser) agree on dimensions and components, the libjpeg builds also on the coefficients. **Display (E18).** Their recorded Pillow, OpenCV and Chrome hashes equal Dec*'s; freshly decoded, 400 accepted files are bit-exact for the AS9 family (3,600/3,600 pairs) and mozjpeg, and 399 for headless Chrome (one file above 4 MP skipped), while IJG 7 and later differ on every 4:2:0 file ($|\Delta| \leq 101$), 9a+ also by 1 on 4:4:4, and non-default settings and Go differ (§R8). **E10.** A Besu QBFT devnet ($n = 4, 7$; 2-s blocks) ran on one host with an *assumed* 5–320-ms inter-bloc round-trip matrix; its queueing models are descriptive (three of six knee predictions failed). All 87 hash-checked runs had 0 block-hash mismatches. A first claim is included in a median 1.90 s (95th percentile 2.83 s; $n = 7$, 25 tx/s, the knee). With $f + 1$ validators stopped the chain halted; the beacon was predictable in 30/30 epochs without pulses and 0/30 with them. A lagging node trailed by up to 110 s under overload and 442 s during a fault, so receipt-only parties saw "status unknown" (G2). Fork tests: §R10.

E11: why a ledger. One attack harness runs over a platform database (A), a CT-style log with gossip (B), a witness-cosigned log (C), a log whose witnesses validate every transition (C′) and the CUTLINE ledger (D), a QBFT model checked against the contracts on EDR (75/75 cells) and on Besu (Table R15). Every H7 clause holds: A fails scenarios Sc1–Sc4; B detects equivocation only after gossip (median 4.7 s at a 10-s period); C′ and D resist Sc1–Sc4 with at most f colluders; C′ equals D except for the court (Sec. VII). 242 of 1,674 cells differ from the frozen prediction, so the model is descriptive; on Besu with $k \leq f$ censoring proposers, D's median inclusion delay was 1.64–1.92 s (Table II).

R17.7 Performance and storage (E12)

Single-thread medians over 2–10 masters per cell (0.5–12 MP; quiet rerun; Table R18): the fast path, which runs the Compile in numpy and re-encodes with libjpeg-turbo, takes 236–307 ms per listing megapixel against a design target of ≤ 20 ms. No native verifier was built. For a 1024-px listing from a 1-MP master, the Python X0 verifier takes 359 ms, one hop 108 ms, the bundle 903 KiB and a capture commit 0.17 s. Profiles (E16 under load; E16b on a quiet host; Table R20) put 74–88% of the fast path in the numpy Compile; the libjpeg-turbo re-encode takes 17.1–19.3 ms/MP on the quiet host, 5–9% of the fast path. This indicates the order of magnitude of a native verifier's encoding step; it does not measure a native verifier. The fast path costs about 49–70 CPU-hours per 10^6 listings. The 12-MP capture commit (9.8 s) is dominated by OpenCV's region dilation, whose 12-thread pool is throttled under the one-core quota of the runs; pinned to one thread it takes 2.95 s (E16b).

R18 Run integrity

Every job ran in its own memory-capped unit on one 12-core, 31-GB host without a usable GPU. Some jobs were killed at their cap and rerun, and one repository-synchronisation incident overwrote result lines of running E4 jobs; the restartable runners recomputed every lost line under the same code digest and pre-registration hash, and no result changed. Per-job caps, peaks and the incident log are in the exit reports of the artifact.

Evaluation-code fixes after the E5 freeze. Three fixes changed frozen evaluation files without changing their semantics, each under a new code digest and applied only after all other runners had finished: the WF3 hop runner reads the source level from the E5 level records (600/600 cross-check); key parsing accepts dataset identifiers containing #; and BiRefNet-lite runs with the ONNX Runtime memory arena off (byte-identical numerics).

E18 reruns. The decoder-setting grid was rerun once because the decoding library ignored the requested settings, and one run that exceeded its memory cap was restarted (§R8, run notes); the affected lines were discarded and recomputed.

R-A Evidence files

Every number in the manuscript, the supplement and this report is taken from the experiment reports of the artifact (`experiments/*/*_exit.md` and the reports they cite); the claim register `paper/OUTLINE.md` maps each printed number to its report. Table R23 lists the generator or result directory behind each figure and generated table. The supplement figures are re-styled by `paper/figs/src/supp_restyle.py`, which runs each generator unchanged and checks its recorded data (`--check`).

TABLE R23. Evidence files behind the figures and tables of the supplement and of this report (paths relative to the root of the project's research repository).

Item	Generator / evidence
Fig. R1 (E1)	`experiments/s5/figs/fig4_e1.py`; data `experiments/s5/e1`
Fig. R2 (E2)	`experiments/s5/figs/fig5_e2.py`; data `experiments/s5/e2`
Figs. R3, R4 (E3)	`experiments/s5/figs/fig6_e3.py`, `fig7_e3.py`; data `experiments/s5/e3`
Fig. R5 (E13)	`experiments/r1/figs/fig_x1_residual.py` (`--check` passes)
Fig. R6, Table R8 (E5)	`experiments/s9/figs/fig9_e5.py`; manifest `experiments/s9/figs/fig9_manifest.json`
Table R10 (E5 hand audits)	labels `experiments/s9/hand_audit/labels_audit1.json`, `labels_audit2.json`; scorer `score_hand_audit.py`; result `hand_audit_result.json`
Figs. R7, R8 (E6)	`experiments/s5/figs/fig8_e6.py`; data `experiments/s5/e6_postfix`, `experiments/s5/e6`
Fig. R9 (E10)	`experiments/s6/e10/fig10.py`; data `experiments/s6/e10`
Figs. R10, R11; Tables R15, R16, R17 (E11)	`experiments/s7/make_figs.py`, `experiments/s7/s7_exit.py`; data `experiments/s7`
Table R3 (E1)	`experiments/s5/figs/tab7_zoo.py`
Tables R4, R21	`experiments/s8/make_tabs.py`
Table R13 (E9, E17, E17b)	`experiments/s6/e10/tab4.py` from `experiments/s6/e9/e9.jsonl`, `experiments/s6/e10/e10.jsonl`, `experiments/r1/x5/x5_compare.json`, `experiments/r1/x5/x5b.jsonl`
Table R18 (E12)	`code/eval/runners/e12.py`, `experiments/s9/figs/tab6_costs.py`
Generated tables (Markdown twins)	`paper/tables/*.md`
Per-job memory caps, peaks and the incident log (§R18)	the exit reports `experiments/*/*_exit.md`

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